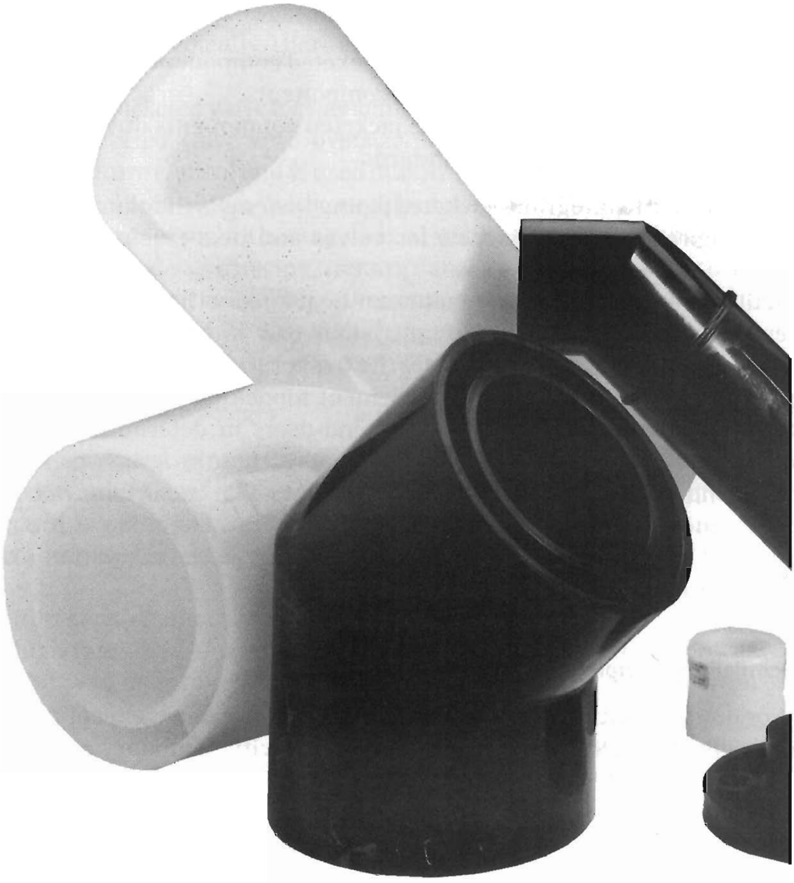


Figure 5. Polypropylene dual-containment piping.

Black polypropylene is preferred as it is resistant to weathering and can be safely used in exposed situations. It is important that pipe and fitting brands should not be mixed.

Polypropylene dual-containment piping systems can also be particularly useful for conveying chemical waste in underground systems under gravity flow conditions.

Typically, the method of jointing is to use primary and secondary couplings, each manufactured with an integral chrome/nickel/alloy resistance wire moulded in place. The wire is electrically heated using a microprocessor-controlled fusion unit that enables uniform jointing to be made in minutes.

Both primary and secondary joints can be assembled in the trench or above ground local to the trench, depending on the site conditions. Polypropylene pipe is characterised by its outstanding chemical resistance, high thermal

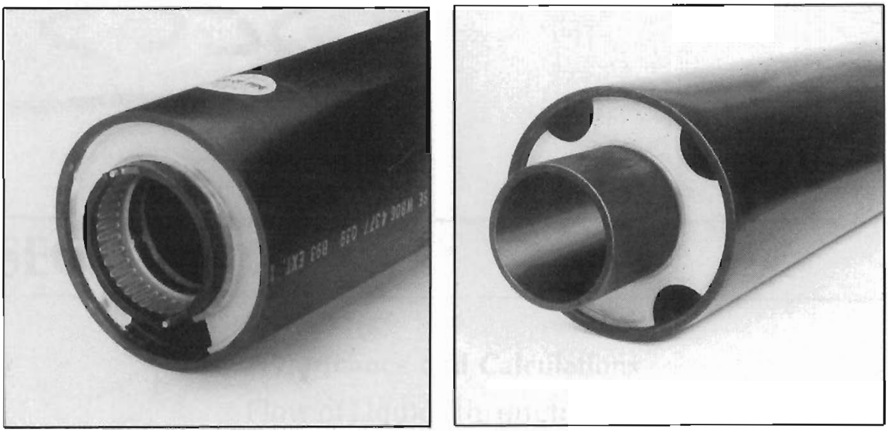


Figure 6. Underground double-containment acid-waste piping.

resistance and good fatigue strength. Polyvinylidene fluoride (PVDF) is resistant to most inorganic acids as well as to alphatic and aromatic hydrocarbons, organic acids, alcohols and halogenated solvents. It is not resistant to alkaline amines, alkalis and alkaline metals. Strongly polarised solvents, such as ketones and organic acid esters, swell PVDF slightly. It has an exceptional resistance to UV radiation and a wide pressure/temperature range.

SECTION 7

Performance and Calculations

Flow of Liquids through Pipes

Liquids pumped or discharged through pipes under conventional pressure behave as incompressible fluids. Flow is assumed to fill the pipe section: basic flow rate (Q) and flow velocity (V) are directly related, viz:

$$Q = V \times \text{pipe bore area (in consistent units)}$$

Flow velocity

Flow velocity is defined as the mean or average velocity at a given cross-section. Due to frictional effects (and the fact that all real fluids possess viscosity), a velocity gradient will exist across a pipe section, ranging from zero at the point of contact with the pipe wall to a maximum at the centre line (Figure 1). The actual velocity profile may be smooth or irregular, depending on whether the flow is laminar or turbulent, respectively (see the section Laminar and turbulent flow below).

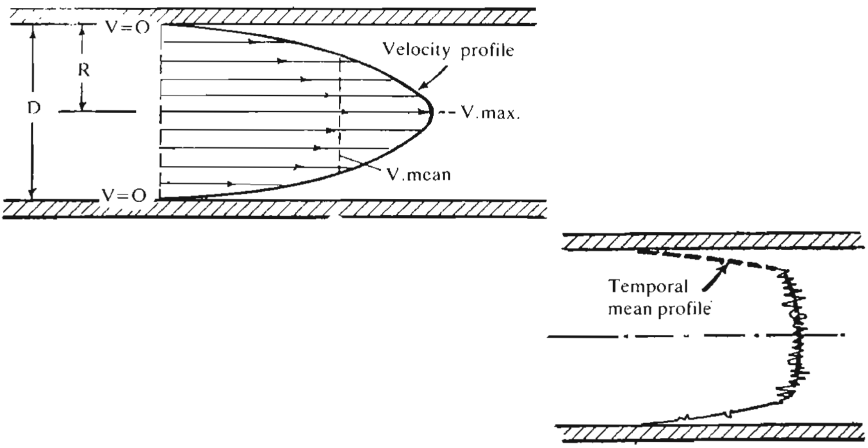


Figure 1.

The basic formula relating Q, V and pipeline diameter (d) is:

$$Q = \frac{\pi}{4} \times V \times d^2$$

In engineering units this becomes:

$$Q = kV \times d^2$$

where k is a constant, depending on the units employed (see Table 1).

Transposed forms of this equation are also useful for direct solutions for V and d. viz:

$$V = \frac{Q}{k \times d^2} \qquad d = \sqrt{\frac{Q}{kV}}$$

Flow velocity V is not necessarily a significant parameter except that it governs frictional losses. For general design work, arbitrary flow-velocity limits are normally assumed, e.g. for water supplies normal design flow velocities are:

General services	1.2 to 3 m/sec (4 to 10 ft/sec)	Gallon = 277 in^3
Water supplies	up to 2 m/sec (up to 7 ft/sec)	One day = 86,400 sec
Boiler feed	2.5 to 4.5 m/sec (8 to 15 ft/sec)	Barrel = 35 Imp gal/min
		42 US gal/min
		m^3 = 6.28 barrels

Table 1. Flow velocity by calculation

Flow rate (Q) unit	Flow velocity		Pipe bore (d) unit
	ft/sec	m/sec	
Cubic in/sec	$Q/10d^2$	$0.03\ Q/d^2$	in
		$20.8\ Q/d^2$	mm
Cubic in/min	$0.00177\ Q/d^2$	$0.00054\ Q/d^2$	in
		$0.348\ Q/d^2$	mm
Gallons per minute	$0.49\ Q/d^2$	$0.15\ Q/d^2$	in
		$97\ Q/d^2$	mm
Litres per minute		$2\ Q/d^2$	in
		$1275\ Q/d^2$	mm
US gallons per minute	$0.408\ Q/d^2$	$0.1250\ Q/d^2$	in
		$80\ Q/d^2$	mm
Tons of water per day	$49\ Q/d^2$	$15\ Q/d^2$	in
Cubic metres of water per day		$14.75\ Q/d^2$	mm
US barrels per day	$0.012\ Q/d^2$	$0.0036\ Q/d^2$	in
		$2.35\ Q/d^2$	mm

Example:

7 in pipe: 0.267 ft^2, 1000 gal/min,
16.7 gal/sec/6.24 = 2.67 cusec/0.267 = 10 ft/sec

$$\text{Vel} = 10\ \text{ft/sec} = \frac{0.49}{49} \times 1000\ \text{gal/min} = 10\ \text{ft/sec}$$

Coefficient = 0.49 — d^2 = 1 million
1 day = 86,400 sec — Area = 0.785 m^2
Pipe 1000 mm diameter

1 million m^3/day/86,400 = 11.55 m^3/sec/0.785 = 14.75 m/sec
$\pi/4 = 0.785\ (1\ \text{Mm}^3/d^2) \times 14.75$ — Pipe 1000 mm dia = d.

Pipe sizing

Rather greater attention to limiting flow velocities is normally required on the suction side of pumps. Also, frictional losses are proportional to fluid viscosity as well as flow velocity. More specific recommendations for flow velocities are given in Tables 2A and 2B. Again, these are largely arbitrary figures based on providing suitable hydraulic conditions in suction pipes or generally acceptable levels of friction loss in delivery pipes (see also Table 3).

Accepting arbitrary valves for flow velocities, the corresponding pipe size (d) required for a specific delivery (Q) follows from simple formula calculation. In the case of water, a general formula often used is:

$$\text{pipe diameter (in)} = \sqrt{\frac{\text{gal/min}}{10}}$$

Table 2A. Recommended suction-flow velocities

Pipe bore		Water		Light oils		Boiling liquids		Viscous liquids	
mm	in	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec
25	1	0.50	1.5	0.50	1.5	0.300	1.0	0.300	1.0
50	2	0.50	1.6	0.50	1.5	0.300	1.0	0.330	1.1
75	3	0.50	1.7	0.50	1.6	0.300	1.0	0.375	1.2
100	4	0.55	1.8	0.55	1.8	0.300	1.0	0.400	1.3
150	6	0.60	2.0	0.60	2.0	0.350	1.1	0.425	1.4
200	8	0.75	2.5	0.70	2.3	0.375	1.2	0.450	1.5
250	10	0.90	3.0	0.90	3.0	0.450	1.5	0.500	1.7
300	12	1.40	4.5	0.90	3.0	0.450	1.5	0.500	1.7
	over 12*	1.50	5.0						

*General formula: pipe diameter (in) = $\sqrt{\frac{\text{gal/min}}{10}}$.

In high-pressure systems, e.g. hydraulic circuits using small-bore pipes, pipe sizing is more critical and normally determined directly from a specified or nominal figure for pressure drop. This involves working an appropriate pressure-drop formula as a solution for pipe bore.

The same technique may also be applied to fluid transport systems, especially if high pressures are involved or working conditions are critical. In this case,

Table 2B. Recommended delivery-flow velocities

Pipe bore		Water		Light oils		Boiling liquids		Viscous liquids	
mm	in	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec
25	1	1.00	3.5	1.00	3.5	1.00	3.5	1.00	3.5
50	2	1.10	3.6	1.10	3.6	1.10	3.6	1.10	3.6
75	3	1.15	3.8	1.15	3.8	1.15	3.8	1.10	3.7
100	4	1.25	4.0	1.25	4.0	1.25	4.0	1.15	3.8
150	6	1.50	4.7	1.50	4.7	1.50	4.7	1.20	3.9
200	8	1.75	5.5	1.75	5.5	1.75	5.5	1.20	4.0
250	10	2.00	6.5	2.00	6.5	2.00	6.5	1.30	4.5
300	12	2.65	8.5	2.00	6.5	2.00	6.5	1.40	4.5
	over 12*	3.00	10.0						

*General formula: pipe diameter (in) $= \sqrt{\dfrac{\text{gal/min}}{20}}$.

Table 3. Pipe bore size for given velocity

Flow velocity		Pipe bore (mm) for flow rate in l/min	Pipe bore (in) for flow rate in gal/min
m/sec	ft/sec		
0.3	1	$8.4\sqrt{Q}$	$0.7\sqrt{Q}$
0.5	—	$6.5\sqrt{Q}$	—
0.6	2	$6.0\sqrt{Q}$	$0.5\sqrt{Q}$
0.9	3	$4.9\sqrt{Q}$	$0.4\sqrt{Q}$
1.0	—	$4.6\sqrt{Q}$	—
1.1	—	$4.4\sqrt{Q}$	—
1.2	4	$4.2\sqrt{Q}$	$0.35\sqrt{Q}$
1.5	5	$3.8\sqrt{Q}$	$0.3\sqrt{Q}$
1.8	6	$3.4\sqrt{Q}$	$0.28\sqrt{Q}$
2.0	—	$3.3\sqrt{Q}$	—
2.1	7	$3.2\sqrt{Q}$	$0.265\sqrt{Q}$
2.4	8	$3.0\sqrt{Q}$	$0.25\sqrt{Q}$
2.5	—	$2.9\sqrt{Q}$	—
2.75	9	$2.8\sqrt{Q}$	$0.23\sqrt{Q}$
3.0	10	$2.6\sqrt{Q}$	$0.22\sqrt{Q}$

since relatively larger pipe sizes are normally involved, recommendations are commonly based on flow rates only, viz:

Suction lines: pressure drop 0.0115 to 0.23 bar per 100 m (0.05 to 1 lbf/in^2 per 100 ft) depending on the available NPSH.

Delivery lines: 0.115 to 1.38 bar per 100 m (0.5 to 6 lbf/in^2 per 100 ft) depending on the flow rate, viz:

(a) 0.46 to 1.4 per 100 m for flow rates up to 450 l/min
(2 to 6 lbf/in^2 per 100 ft for flow rates up to 100 gal/min)
(b) 0.33 to 1.15 bar per 100 m for flow rates from 450 to 900 l/min
(1.5 to 5 lbf/in^2 per 100 ft for flow rates from 100 to 200 gal/min)
(c) 0.23 to 0.92 bar per 1200 m for flow rates from 900 to 2250 l/min
(1 to 4 lbf/in^2 per 100 ft for flow rates from 200 to 500 gal/min)
(d) 0.11 to 0.46 bar per 100 m for flow rates above 2250 l/min
(0.5 to 2 lbf/in^2 per 100 ft for flow rates above 500 gal/min).

Pipe sizing by specific gravity of fluid

The size of delivery lines on centrifugal pumps is sometimes based on economic flow velocity related both to the specific gravity (SG) of the fluid being handled and the type of driver. This is realistic in the sense that the power input required, and thus the cost of pumping, is directly proportional to fluid specific gravity, and economic flow velocity varies inversely to pump speed.

Recommended flow velocities are given in Table 4.

Table 4. Recommended flow velocities based on fluids SG

Pipe diameter		Power-driven pumps						Turbine-driven pumps					
		SG = 1.0		SG = 0.75		SG = 0.5		SG = 1.0		SG = 0.75		SG = 0.5	
mm	in	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec	ft/sec
50	2	1.80	6.00	2.10	7.00	2.30	7.5	1.50	5.00	1.70	5.50	1.80	6.00
75	3	2.10	7.00	2.40	8.00	2.60	8.5	1.70	5.50	1.80	6.00	2.00	6.50
100	4	2.40	8.00	2.75	9.00	3.00	10.0	1.80	6.00	2.00	6.50	2.15	7.00
150	6	2.75	9.00	3.00	10.00	3.65	12.0	2.00	6.50	2.15	7.00	2.40	8.00
200	8	3.00	10.00	3.40	11.25	4.00	13.0	2.10	6.75	2.30	7.50	2.60	8.50
250	10	3.25	11.00	3.65	12.00	4.20	14.0	2.15	7.00	2.35	7.75	2.75	9.00
300	12	3.50	11.50	3.80	12.50	4.40	14.5	2.15	7.00	2.40	8.00	2.80	9.25
350	14	3.60	11.75	4.00	13.00	4.50	15.0	2.15	7.00	2.40	8.00	2.90	9.50
400	16	3.65	12.00	4.00	13.00	4.60	15.0	2.15	7.00	2.40	8.00	2.90	9.50
						and over							

Critical flow velocity

Flow velocity is 'critical' in the sense that it is a major factor in determining the frictional losses of the flow. This is not necessarily significant for fluid transport applications involving flow velocities within the recommended ranges.

Flow velocity can, however, be a critical factor in practical applications involving the transport of solids in suspension in a fluid. The flow velocity will largely govern whether the solids are transported in suspension (homogeneous flow), or whether the solids tend to settle out forming sliding layers over a settled bed (heterogeneous flow).

To produce homogeneous flow it is necessary that the flow velocity should be greater than the fall velocity of the solids in the fluid. This sets critical or minimum flow-velocity requirements for the handling of fluids containing solids in suspension, which can only be determined satisfactorily on empirical lines. Some specific recommendations are given in Table 5.

See also the chapter on Flow of Mixtures through Pipes.

Laminar and turbulent flow

Flow through pipes can be either laminar or turbulent, the flow condition being significant in affecting both the velocity gradient and the frictional losses. With laminar flow, frictional losses are due to viscous drag and are independent of the condition of the pipe bore. With turbulent flow, viscous shear forces predominate and the condition of the boundary surface can materially affect the total friction.

The actual flow condition can be established by reference to a non-dimensional parameter, the Reynolds number (R_e), determined as:

$$R_e = \frac{dV}{\nu}$$

where d = pipe bore
V = velocity of flow
ν = kinematic viscosity of the fluid (in consistent units)

Note: The Reynolds number itself is dimensionless.

Table 5. Minimum flow velocities for slurries

Type	Size of solids (mesh no.)	Flow velocity	
		ft/sec	m/sec
Fines	Over 200	3–5	1.00–1.50
Sands	200–20	5–7	1.50–2.00
Coarse	20–4	7–11	2.00–3.25
Sludge		11–14	3.25–4.25

In engineering units:

$$R_e = \frac{7740\,dV}{\nu} \qquad \text{or} \qquad R_e = \frac{930\,dV}{\nu}$$

where d is in inches, V is in ft/sec, ν is in centistokes — where d is in cm, V is in m/sec, ν is in centistokes

In the case of clean cold water:

$$R_e = 7740\ dV \text{ when d is in inches, V in ft/sec}$$
$$= 930\ dV \text{ when d is in cm, V in m/sec.}$$

Also, see Table 6.

The same formulae apply for the calculation of Reynolds numbers for flow in non-circular pipes, substituting the equivalent hydraulic diameter for the circular diameter, viz:

$$\text{Equivalent hydraulic diameter} = 4 \times \left(\frac{\text{cross-sectional flow area}}{\text{wetted parameter}}\right)$$

The quantity contained in the brackets is the *hydraulic radius* of a non-circular pipe.

The flow is laminar at Reynolds numbers up to 2000; and turbulent at Reynolds numbers above approximately 4000. In the transitional range (R_e = 2000–4000), flow can vary from laminar to turbulent and flow conditions are indeterminate.

In the case of laminar flow the velocity gradient will be linear. Maximum velocity at the centre of the bore will be of the order of 1.5 times the mean flow velocity.

With turbulent flow there is no clearly defined velocity profile. The temporal mean profile will be of the form shown in Figure 2, the actual profile varying with the Reynolds number. At low Reynolds numbers the maximum flow velocity (at the centre of the bore) will be of the order of 2.0 times the mean velocity, reducing to about 1.25 times the mean velocity at higher Reynolds numbers.

Table 6. Reynolds number for clean cold water

	Pipe bore									
mm (in)	25 (1)	40 ($1^1/_2$)	50 (2)	75 (3)	100 (4)	150 (6)	200 (8)	250 (10)	300 (12)	450 (18)
Per 1 l/min*	835	550	420	280	210	140	105	85	70	46
Per 1 gal/min*	3800	2500	1900	1270	950	630	475	380	320	210

*Multiply by actual numerical value in either unit to give Reynolds number.

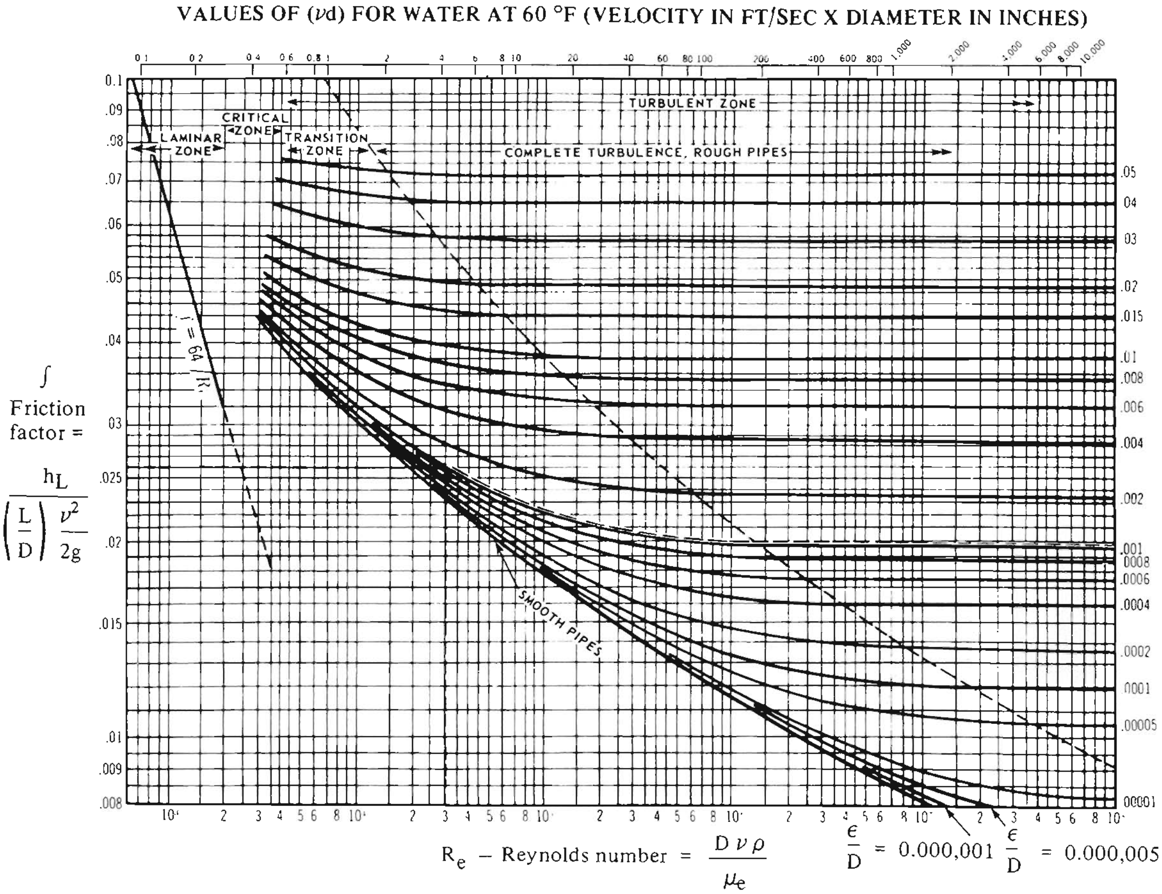


Figure 2. Friction factors for pipes. Example: friction factor for pipe with relative roughness 0.001 at flow Reynolds number of 30,000 = 0.026.

The velocity profile is primarily of significance where a pilot tube or similar flow-measuring device is inserted in the pipe as this will be subject to position error. In the cross-section with turbulent flow there is no point where the local velocity is likely to be constant and fully predictable. This factor is not necessarily significant where measurements or calculations are based on flow rate, or mean flow velocity, the latter being determined directly from the flow rate and pipe bore.

Frictional losses

Frictional losses are calculated in terms of pressure drop or, alternatively, head loss. Figures for frictional losses are normally reduced to (friction) head equivalent per m (ft) or per 100 m (ft) of pipe and are then directly applicable to any length or aggregate length of straight run of pipe of the sizes concerned. Such data are available and presented both in graphical and tabular form for a wide range of pipe sizes, for water, oils and fluids. Agreement is not always good between such data originating from different sources. Many, particularly for water flow through pipes, are based on formulae more than half a century old and have a limited range of accuracy. Others are based on quite widely differing empirical coefficients, with similar limitations. If the reliability of the data available is suspect, or shows inconsistencies, more accurate solutions will be arrived at by working from basic principles.

In the case of laminar flow, the D'Arcy–Weisbach formula can be applied in the form:

$$\Delta p = f\frac{L\rho V^2}{2\,Dg}$$

where ΔP = pressure drop in $\mathrm{lbf/in^2}$
L = length of pipe in feet
V = flow velocity
D = bore of pipe in inches
ρ = mass density of fluid
f = a friction factor = $\dfrac{64}{\text{Reynolds number}}$

In the case of larger pipes (e.g. 25 mm (1 in) bore and above), and expressed in terms of flow rate (Q) rather than flow velocity (V), the following simplified formulae can be used:

$$\Delta P = f\frac{Q^2 L}{K_1 D^5} \times \text{specific gravity of fluid}$$

where K_1 is a constant dependent on the units adopted for Q, L and D.

The corresponding formula for head loss (ΔH) is:

$$\Delta H = f\frac{Q^2L}{K_2D^5}$$

It must be noted that before these formulae can be used the Reynolds number must be determined to:

(i) Establish that the flow is laminar ($R_e \leq 2000$)

(ii) Calculate the friction factor for the formula $\left(f = \frac{64}{R_e}\right)$.

With laminar flow, friction loss (pressure drop or head loss) is not affected by the roughness of the pipe bore (unless this appreciably modifies the effective bore size). The formulae are thus directly applicable to laminar flow in all types, construction and ages of circular pipes and non-circular pipes (the latter with R_e computed on the basis of their hydraulic radius).

Turbulent flow

In a majority of practical cases of pumped fluids, flow is turbulent (i.e. $R_e \geq 4000$) and simple calculation of flow losses no longer applies. Basically, the D'Arcy formula can be used, but with a different friction factor (f_{turb}), the value of which is dependent both on the Reynolds number and the surface roughness of the pipe bore.

Specifically, when the Reynolds number of flow exceeds 2000 there is a critical zone into which laminar flow may extend (but flow conditions are unpredictable and may change from laminar to turbulent, and vice versa), followed by a region of developing turbulent flow where the friction factor decreases with increasing Reynolds number but increases with increasing bore-surface roughness. Finally, full turbulent flow is established, when the friction factor is a constant regardless of increase in the Reynolds number and is dependent only on surface roughness.

For smooth-bore pipes, the following formulae can be used to calculate the friction factor directly:

$$f_{turb} = \frac{0.3164}{{R_e}^{0.25}} \quad \text{for } R_e \text{ values between 4000 and 10,000}$$

$$f_{turb} = 0.0032 + \frac{0.221}{{R_e}^{0.237}} \quad \text{for } R_e \text{ values over 10,000}$$

Working data for friction factors for turbulent flow are usually presented in chart form (Figure 3) where these zones are clearly seen. Effectively, turbulent flow friction factors are bonded by a lower curve, representing the friction factor for smooth-bore pipes, surmounted by a series of curves for pipes with increasing surface roughness. Roughness is defined in terms of relative roughness or ϵ/D.

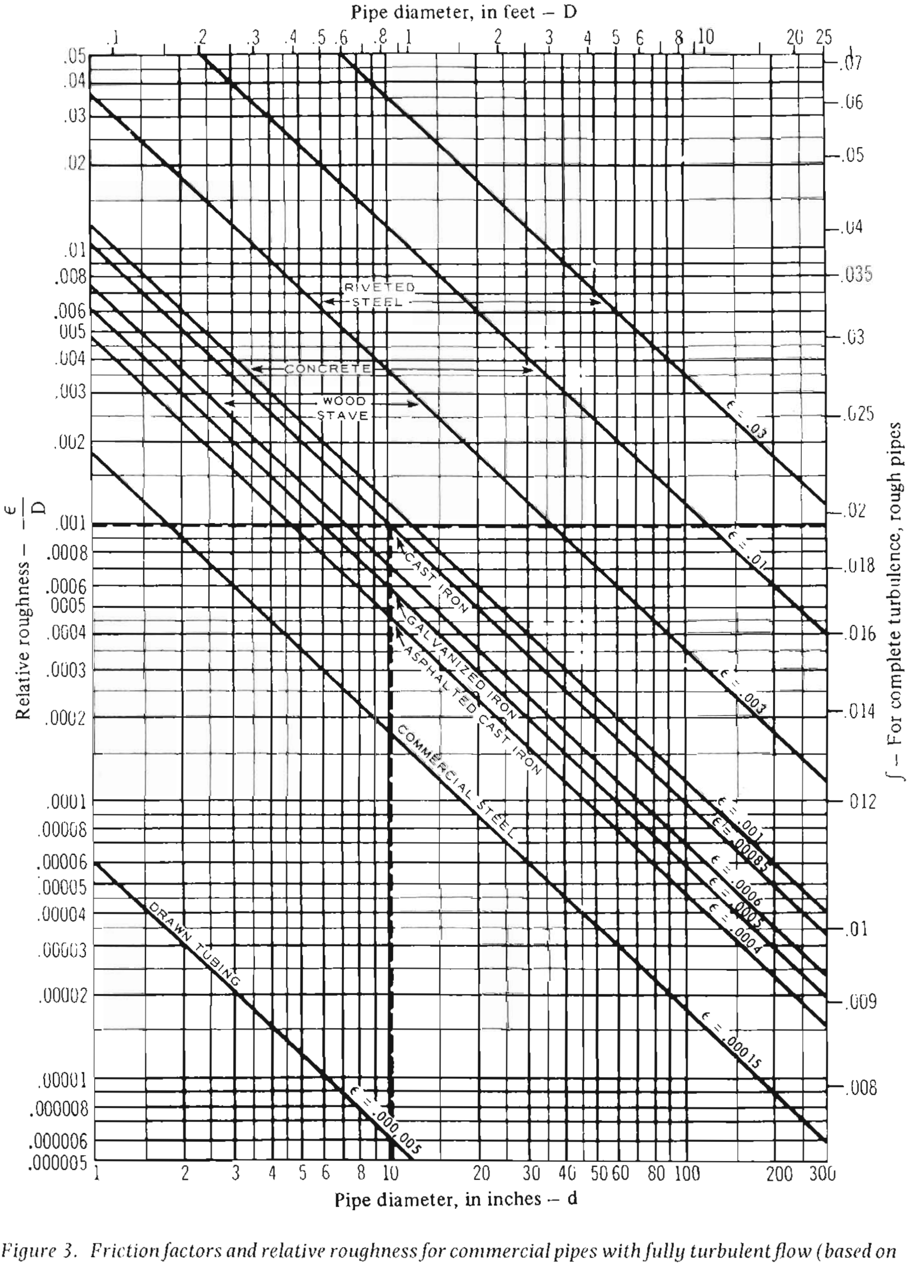


Figure 3. Friction factors and relative roughness for commercial pipes with fully turbulent flow (based on ASME data originated by L. F. Moody). Example: for 10 in diameter cast-iron pipe, relative roughness (ϵ/D) = 0.00085. Friction factor = 0.0196.

where ϵ is absolute roughness, or effective height of pipe-wall irregularities, and D the pipe bore, in the same units.

An alternative form of chart is shown in Figure 4, together with typical values of ϵ for different pipe materials, from which factors for turbulent flow can be read directly.

It must be appreciated that the relative roughness of pipes is increased by corrosion, and also by age, due to encrustations. Data given in Figures 3 and 4 apply to new pipes in clean condition.

Basic formulae then applicable are:

$$\frac{\Delta P}{L} = f_{turb} \times \frac{\rho V^2}{2Dg}$$

$$\frac{\Delta HP}{L} = f_{turb} \times \frac{\rho V^2}{2D\omega}$$

where ρ is the fluid density
ω is the fluid specific weight.

Practical formulae for calculations:

Reynolds number (R_e):

$$R_e = 6.31\frac{W}{d\mu}$$

for W in lb/hr
d in inches
μ in centipoise

$$R_e = 0.482\frac{Q \times SG}{d\mu}$$

for Q in ft^3/hr
d in inches
μ in centipoise

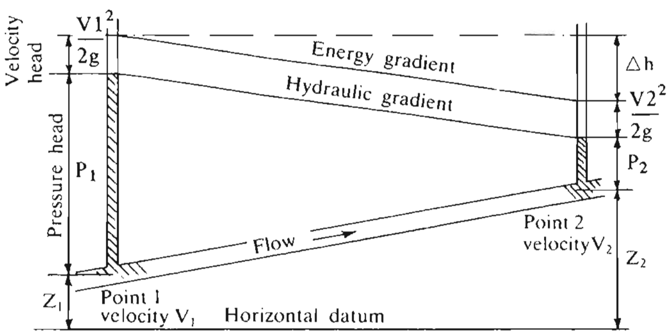


Figure 4. Simplified energy and pressure gradients.

$$R_e = \frac{dV}{v}$$

for d in ft
V in ft/min
v in ft^2/sec

$$R_e = 7740\frac{dV}{v}$$

for d in inches
V in ft/sec
v in centistokes

$$R_e = 1{,}419{,}000\frac{Q}{dv}$$

for Q in ft^3/sec
d in inches
v in centistokes

$$R_e = 3160\frac{Q}{dv}$$

for Q in ft^3/hr
d in inches
v in centistokes

$$R_e = 394\frac{W\bar{V}}{dv}$$

for W in lb/hr
$\bar{V}$ (specific volume) in ft^3/lb
d in inches
v in centistokes

$$R_e = 92.9\frac{dV}{v}$$

for d in millimetres
V in mm/sec
v in centistokes

$$R_e = 2274\frac{Q}{dv}$$

for Q in mm^3/hr
d in mm
v in centistokes

Flow velocity (V):

$$V = 183.3\frac{Q}{d^2}$$

for Q in ft^3/sec
d in inches

$$V = 0.408\frac{Q}{d^2}$$

for Q in US gal/min
d in inches

$$V = 0.00389\frac{Q \times SG}{\rho d^2}$$

for Q in ft^3/hr
ρ in lb/ft^3
d in inches
SG = specific gravity

$$V = 0.340\frac{Q}{d^2}$$

for Q in Imperial gal/min

d in inches

$$V = 3350 \frac{Q}{d^2}$$

for Q in m^3/hr
d in mm

Head loss (laminar flow):

$$H_L \text{ (ft)} = 0.0962 \frac{LV\mu}{d^2 \rho}$$

for L in ft
V in ft/sec
μ in centipoise
d in inches
ρ in lb/ft^3

$$H_L \text{ (ft)} = 0.0393 \frac{LQ\mu}{d^4 \rho}$$

for L in ft
Q in US gal/min
μ in centipoise
d in inches
ρ in lb/ft^3

$$H_L \text{ (ft)} = 0.0049 \frac{LW\mu}{d^4 \rho^2}$$

for L in ft
W in lb/hr
μ in centipoise
d in inches
ρ in lb/ft^3

$$H_L \text{ (ft)} = 0.0328 \frac{LQ\mu}{d^4 \rho}$$

for L in ft
Q in Imperial gal/min
μ in centipoise
d in inches
ρ in lb/ft^3

$$H_L \text{ (m)} = 107 \frac{LV\mu}{d^2 \rho}$$

for L in inches
V in mm/sec
μ in centipoise
ρ in tonnes/m^3
d in milli-inches

$$H_L \text{ (m)} = 2670 \frac{LQ\mu}{d^4 \rho}$$

for L in inches
Q in mm^3/bar
d in mm
ρ in tonnes/m^3

Pressure drop (laminar flow):

$$\Delta P \text{ (lb/in}^2\text{)} = 0.000668 \frac{LV\mu}{d^2}$$

for L in ft
V in ft/sec
μ in centipoise
d in inches

$$\Delta P \text{ (lb/in}^2\text{)} = 0.1225 \frac{LQ\mu}{d^4}$$

for L in ft
Q in ft^3/sec
μ in centipoise
d in inches

$$\Delta P\ (\text{lb/in}^2) = 0.000034 \frac{LW\mu}{d^4\rho}$$

for L in ft
W in lb/hr
μ in centipoise
d in inches
ρ in lb/ft^3

$$\Delta P\ (\text{lb/in}^2) = 0.000273 \frac{LQ\mu}{d^4}$$

for L in ft
Q in US gal/min
μ in centipoise
d in inches

$$\Delta P\ (\text{lb/in}^2) = 0.0002275 \frac{LQ\mu}{d^4}$$

for L in ft
Q in Imperial gal/min
d in inches

$$\Delta P\ (\text{bar}) = 0.030 \frac{LV\mu}{d^2}$$

for L in inches
W in in/sec
d in milli-inches

$$\Delta P\ (\text{bar}) = 9.05 \frac{LQ\mu}{d^4}$$

for L in inches
Q in l/min
d in milli-inches

Head loss (turbulent flow):

$$H_L\ (\text{ft}) = 0.1863\, f \frac{LV^2}{d}$$

where f = friction factor
L is in ft
V is in ft/sec
d is in inches

$$H_L\ (\text{ft}) = 6260\, f \frac{LQ^2}{d^5}$$

where f = friction factor
L is in ft
Q is in $\text{ft}^3\text{/sec}$
d is in inches

$$H_L\ (\text{ft}) = 0.0311\, f \frac{LQ^2}{d}$$

where f = friction factor
L is in ft
Q is in US gal/min
d is in inches

$$H_L\ (\text{ft}) = 0.026\, f \frac{LQ^2}{d^5}$$

where f = friction factor
L is in ft
Q is in Imperial gal/min
d is in inches

$$H_L\ (\text{ft}) = 0.000483\,f\,\frac{LW^2\bar{V}^2}{d^5}$$

where f = friction factor
L is in ft
W is in lb/hr
$\bar{V}$ (specific volume) is in ft^3/lb
d is in inches

$$H_L\ (\text{ft}) = 0.01524\,f\,\frac{LB^2}{d^5}$$

where f = friction factor
L is in ft
B is in barrels (42 US gal)/hr
d is in inches
d is in inches

$$H_L\ (\text{m}) = 0.041\,f\,\frac{LV^2}{d}$$

where f = friction factor
L is in m
Q is in m/sec
d is in mm

$$H_L\ (\text{m}) = 641{,}270\,f\,\frac{LQ^2}{d^5}$$

where f = friction factor
L is in m
Q is in l/min
d is in mm

Note: These formulae may be used for both laminar flow and turbulent flow, with appropriate friction factors.

Pressure drop (turbulent flow):

$$\Delta P\ (\text{lbf/in}^2) = 0.001294\,f\,\frac{L\rho V^2}{d}$$

where f = friction factor
L is in ft
ρ is in lb/ft^3
V is in ft/sec
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 4315\,f\,\frac{L\rho Q^2}{d^5}$$

where f = friction factor
L is in ft
ρ is in lb/ft^3
Q is in ft^3/sec
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.000216\,f\,\frac{L\rho Q^2}{d^5}$$

where f = friction factor
L is in ft
ρ is in lb/ft^3
Q is in US gal/min
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.00018\,f\,\frac{L\rho Q^2}{d^5}$$

where f = friction factor
L is in ft
ρ is in lb/ft^3
Q is in Imperial gal/min
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.00000336\,f\,\frac{LW^2\bar{V}}{d^5}$$

where f = friction factor
L is in ft

$$\Delta P\ (\text{lbf/in}^2) = 0.0001058\,f\,\frac{L\rho B^2}{d^5}$$

where f = friction factor
L is in ft

W is in lb/hr
$\bar{V}$ (specific volume) is in ft^3/lb
d is in inches

ρ is in lb/hr
B is in barrels (42 US gal) ph
d is in inches

$$\Delta P\ (\text{bar}) = 0.000001125\, f\, \frac{L\rho V^2}{d}$$

where f = friction factor
L is in m
V is in m/sec
d is in mm
ρ is in tonnes/m^3

$$\Delta P\ (\text{bar}) = 0.1613\, f\, \frac{L\rho Q^2}{d^5}$$

where f = friction factor
L is in m
ρ is in tonnes/m^3
Q is in l/mm
d is in milli-inches

Note: These formulae may be used for both laminar flow and turbulent flow, with appropriate friction factors.

Other formulae

Other charts or friction factor data may show substantially different values of friction factor for similar values of relative roughness. This is because the general formula is one of the form:

$$\frac{\Delta H}{L} = f \times \frac{Q^2}{kD^5}$$

which includes the Reynolds number as a factor. The value of the friction factor (f) derived is thus adjusted accordingly.

Colebrook–White equation

The Colebrook–White equation for transitional flow is extensively used for determining the hydraulic performance of sewage, drainage and effluent systems.

The general form of the equation is:

$$\frac{1}{\sqrt{\lambda}} = -2\log_{10}\left(\frac{k_s}{3.7D} + \frac{2.51}{R_e\sqrt{\lambda}}\right)$$

where λ = friction coefficient $\dfrac{2gDi}{V^2}$

k_s = linear measure of effective roughness (m)
D = pipe internal diameter (m)
R_e = Reynolds number, $\dfrac{VD}{\nu}$

The equation expressed in engineering terms is:

$$V = -2\sqrt{(2gDi} \log_{10} \frac{k_s}{3.7D} + \left(\frac{2.51\nu}{D\sqrt{(2gDi}}\right)$$

where V = velocity (m/sec)
g = gravitational acceleration (9.81 m/sec^2)
i = hydraulic gradient
ν = kinematic viscosity of fluid (m^2/sec).

The Hydraulic Research Paper No. 4 recommends values for the linear measure of effective roughness for the commonly used pipeline materials in various conditions. The values appropriate to ductile-iron pipelines are given in Table 7.

Generally, there is no significant deterioration with time of the linear measure of effective roughness k_s where cement mortar-lined or bitumen-lined pipes are conveying treated potable waters. However, conveying certain raw waters can lead to a build-up of slime in the bores of all pipes and this will cause an increase in the value of k_s. The formation of these slime deposits is not deleterious to the linings, and periodic cleaning of this type of main will restore the hydraulic performance virtually to that of the pipeline in its new condition.

The empirical Hazen–Williams formula has the advantage of simplicity and, for determining the flow of raw or potable water at normal temperatures, it can be relied upon to give results of sufficient accuracy for all practical purposes. It is widely used for calculating flow in raw- and potable-water pipelines.

The formula can be conveniently expressed as:

$$V = 0.457 \times 10^{-5} CD^{0.63} i^{0.54}$$

or

$$Q = 0.359 \times 10^{-5} CD^{2.63} i^{0.54}$$

where V = velocity (m/sec)
Q = quantity (l/sec)
D = pipe internal diameter (mm)
i = hydraulic gradient (dimensionless)
C = Hazen–Williams friction coefficient (dimensionless).

By selecting the appropriate value for the coefficient 'C', the Hazen–Williams formula can be used for all types of pipe materials. Table 8 gives values of 'C' for ductile-iron pipelines. The values shown are based on case studies and consequently account is automatically taken of losses due to irregularities at the joints.

The same comments about the linear measure of effective roughness k_s apply equally to the friction coefficient 'C'.

The values of 'C' as given in Table 8 are approximately correct at a velocity of 1 m/sec. At other velocities the approximate corrections are given in Table 9.

Effect of inclined flow

Pressure drop (or equivalent head loss) due to flow is the same in straight pipes whether the pipe run is horizontal, vertical or inclined. With vertical or inclined flow, however, pressure drop is modified by the difference in actual head involved. Here, the Bernoulli theorem applies in defining the total energy at any particular point above any arbitrary horizontal datum plane. Total energy is equal to the sum of the elevation head, the pressure head and the velocity head, i.e.:

$$\text{Total head} = Z + \frac{P}{\rho} + \frac{V^2}{2g}$$

The total head (H) will be a constant for any point. Thus, if the friction loss between points Z_1 and Z_2 on an inclined run (Figure 5) is expressed as a head loss ΔH:

H at point Z_1 = H at point Z2.

Thus:

$$Z_1 + \frac{P_1}{\rho_1} + \frac{V_1{}^2}{2g} = Z_2 + \frac{P_2}{\rho_2} + \frac{V_2{}^2}{2g} + \Delta H$$

Confusion can be caused by reference to hydraulic gradient rather than head or pressure loss. Fundamentally, steady flow conditions through a system can be analysed in terms of a series of Bernoulli equations appropriate to specific lengths of the system, from which may be derived energy curves and pressure curves, as shown simply in Figure 5. The energy curve, normally referred to as the energy gradient, shows the total energy at any point in the system. The pressure curve, normally referred to as the hydraulic gradient, shows the (pressure) head at any point in the system. The energy gradient will always drop in the direction of flow in the discharge side of the energy into potential energy, e.g. at a sudden expansion. Over a section not subject to changes, e.g. a straight length of pipe, the hydraulic gradient effectively represents the friction head loss and may be referred to as such. The term is rarely used in practical engineering calculations, however.

Water hammer

'Water hammer' is the name given to the distinctive 'knocking' noise which can develop in a closed pipe system when the flow velocity is suddenly changed, e.g. by the sudden opening or closure of a tap, valve or other flow-control device. It can also be produced by other factors causing abrupt changes in flow velocity, e.g. sudden starting or stopping of a pump, or abrupt changes in speed of a pump feeding the system.

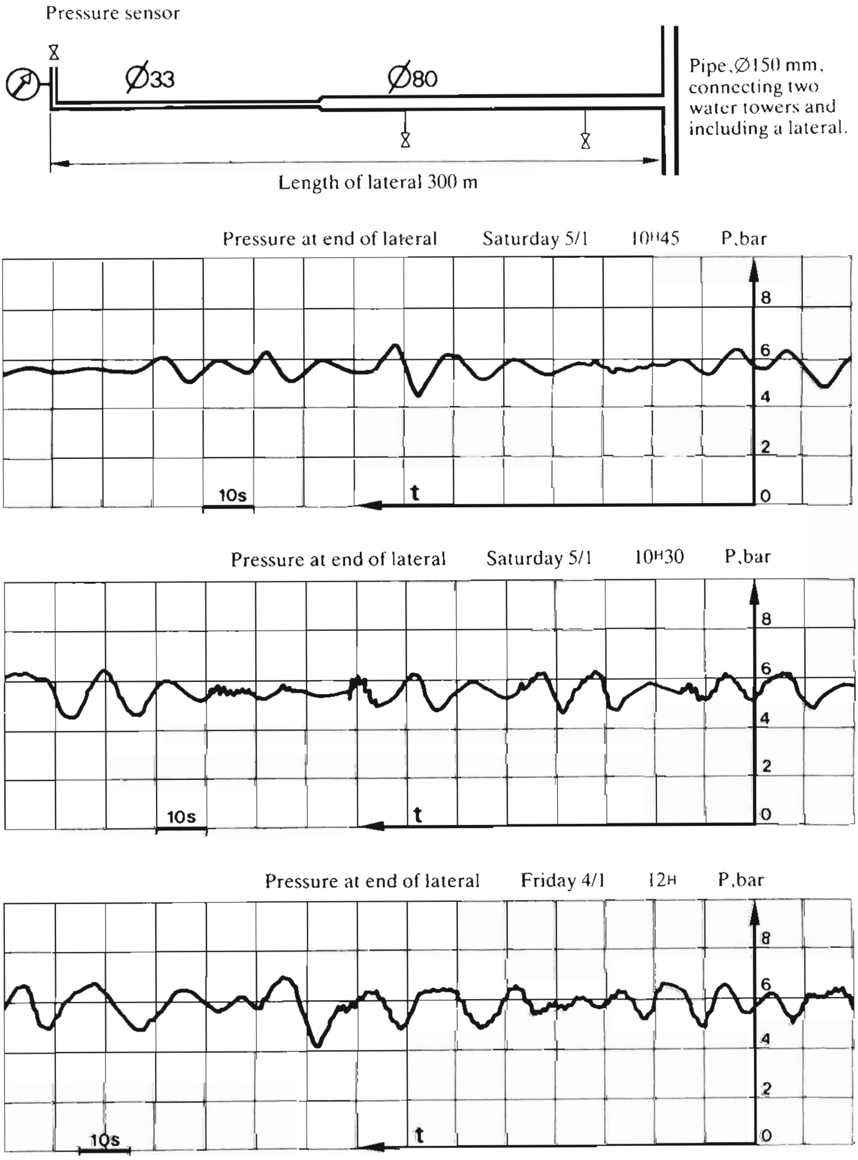


Figure 5. Typical recording of pressure at end of lateral.

The cause of the 'hammer' is a sudden pressure rise in the liquid, caused by the rapid acceleration or deceleration imparted to it, which travels as a pressure wave along the length of pipe, and is reflected backwards and forwards. In addition to generating knocking noises or 'hammer', if the pressure rise is excessive it can cause damage to the piping or system components.

The pressure, velocity and time for the pressure wave to travel from one end of the pipe to the other can be determined from first principles, viz:

$$\text{Velocity of pressure wave (v)} = \frac{4660}{(1 + KD/t)}$$

where D = pipe diameter
t = pipe-wall thickness in same units

$$K = \frac{\text{elastic modulus for liquid}}{\text{elastic modulus for pipe material}}$$

Values of K for water and common pipe materials are:

Pipe material	K
Steel	0.010
Wrought iron	0.0107
Cast iron	0.025
Asbestos/cement	0.088

Pressure rise, expressed as Head (H):

$$H = \frac{V\Delta V}{g}$$

where ΔV = reduction in liquid velocity

Time for pressure wave to travel length L:

$$t = \frac{2L}{\Delta V} \text{ sec}$$

Note that L is the length from the appliance concerned (producing the velocity change) to the ends of the pipe.

There are various methods of reducing the intensity of the pressure wave and thus the shock or degree of hammer. The magnitude of the pressure wave is directly proportional to reduction in flow velocity. so it follows that:

(i) Lowering the flow velocity will be effective, i.e. reducing the flow rate for a given size of pipe or increasing the pipe size for a given flow rate, because ΔV must then be less.

(ii) Decreasing the rate of closure will have a similar effect if the flow stopping time is increased to several times the value of t (times of travel of pressure wave corresponding to instant closure). Surge compressor valves are usually designed on these lines.

Other methods which can provide a cure for water hammer, rather than designing for the system parameters to avoid hammer, are:

(i) air-injection, and
(ii) introducing a flexible element into the system.

Air aspiration is mainly applicable to larger pipelines, the entrained air so supplied acting as a cushion to absorb pressure surges. Air-relief valves can also be installed to relieve air and water during a surge.

Chatter

'Chatter' is rather like water hammer in characteristics but is the result of elasticity in the fluid system. Such elasticity may be introduced by aeration of air entrainment. Under certain conditions, axial oscillation of the fluid column may then develop in the system. In the case of high-pressure systems, chatter may also arise from a mechanical fault, e.g. seal chatter.

Surge-prone pipe systems

Calculation programs using mathematical models can simulate the most characteristic types of phenomena which give rise to pressure surges and can provide a theoretical determination of the range of over-pressures which are likely to affect pipelines.

In practice, while such calculations are available for individual pipes and simple pipe systems, it is difficult to make assumptions covering all possible cases for complex networks. These may contain distribution points, pumps, valves, etc., in which there may be interference, simultaneous or otherwise, between the various situations caused by the different components. Furthermore, the attenuation of such phenomena in both time and space may only be imperfectly known.

The typical recording shown in Figure 6 was made at the end of one of the laterals of a small municipal distribution system.

In view of the complexity of the problem, it may be worthwhile carrying out a diagnosis of the system during operation by locating pressure gauges at various points to detect and record over fairly long periods all the dangerous pressure variations.

The recording can then be analysed in order to determine whether a given sector is subjected to over-pressure or depressurisation problems, and to assess the frequency and importance of such anomalies. Assumptions can be made, by studying the shape of the pressure versus time curves, concerning the type of component or operation which gives rise to such occurrences.

Diagnostic systems have been developed consisting of a suitable number of sensors indicating the values of the various characteristics parameters selected, and a micro-computer acting as a central measurement system.

The micro-computer can take readings from the pressure sensor at a frequency of 1000 readings per second. The other sensors operate at longer time intervals which can be varied according to requirements.

The system has been designed to operate either as a controller or as a central measurement unit. In the latter operating mode, suitable software enables the micro-computer to provide operating personnel with information about the internal behaviour of the system. Operators can thus take measurements at various points in the network and gain vital information enabling them to ascertain:

(i) Unknown sources of possible damage to the pipelines caused by equipment items, and the cause of certain malfunctions.
(ii) Risk of contaminating the mains water supply by ingress of pollution as a result of depressurisation.
(iii) In certain cases, the nuisance factor experienced by consumers owing to pressure surges caused by local users.

Flow of Mixtures through Pipes

Standard formulae for pipeline performance calculations (e.g. pressure drop or head loss) incorporate fluid viscosity as a constant parameter, i.e. as a specific value dependent on the working temperature of the fluid involved. Such calculations are valid only for Newtonian fluids where viscosity remains constant with agitation or change in shear rate. Typical Newtonian fluids include water, aqueous solutions, mineral oils, hydrocarbons, syrups and some resins.

Various other types of fluid are essentially non-Newtonian in characteristics, when the viscosity value under any specific conditions is an apparent one rather than a true one. Such fluids may be categorised as follows:

(i) Fluids containing solids in suspension, further categorised as slurries, sludges and pulps (paper stock).

(ii) Thixotropic fluids, where viscosity decreases as agitation or shear rate is increased. Thixotropic fluids exhibit a hysteresis effect in that their apparent or instantaneous viscosity is dependent on the previous history of the fluid. Figure 1 is a typical rheogram for a thixotropic fluid under laminar flow. Turbulent flow will tend to change the structure of the fluid, which will recover if left standing for a sufficient time. Fluids of this type include greases, soaps, starches, vegetable oils, varnishes, some resins, tars, asphalts, glues and some inks.

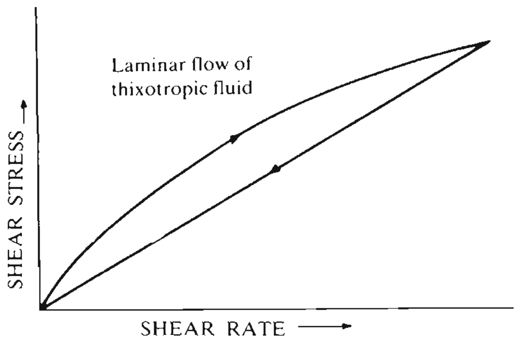


Figure 1.

(iii) Colloidal fluids which behave like thixotropic fluids but will not recover their original viscosity when agitation is stopped. Fluids of this type include colloidal solutions of soaps in water, and oils, lotions, shampoos and gelatinous compounds.
(iv) Dilatent fluids where viscosity increases as agitation or shear rate is increased. Fluids of this type include clays and some slurries.
(v) Rheopectic fluids where viscosity increases with increasing agitation in shear rate up to a maximum value at any constant rate of agitation.
(vi) Plastic and pseudo-plastic fluids where viscosity increases with increasing shear rate, but initial viscosity may be so high as to prevent start of flow in a normal pumping system. These are also known as Bingham fluids.

Strictly speaking, only plastic fluids are true Bingham fluids and include such products as drilling muds, thick mineral slurries and sewage sludge. Pseudo-plastic fluids exhibit a different shear rate–shear stress relationship. Fluids in this category include paper stock, detergent slurries, some paints and lacquers, some mineral slurries, mayonnaise, and cellulose acetate in acetone. A further sub-category of such fluids is known as yield pseudo-plastic, typical products of this type being clay–water suspensions and polymer solutions.

Complex mixture flow

Complex mixture flow may be homogeneous, pseudo-homogeneous, heterogeneous or complex, according to the phase(s) involved and the size of the solids involved (see Figure 2). Homogeneous flow applies only in pure liquid flow. Simple mixtures involve two-phase flows, complex mixtures multi-phase flows. In pseudo-homogeneous flow the solids are present in finely divided,

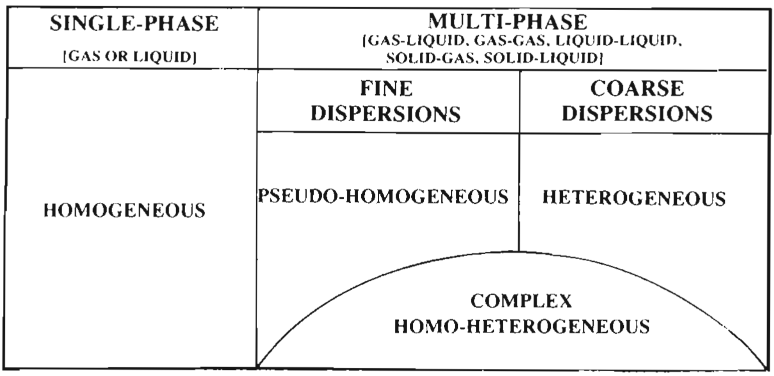


Figure 2. Régimes for homogeneous and heterogeneous flow.

highly dispersed form with almost uniform dispension in the carrier phase. The whole mixture then tends to behave as a single-phase fluid.

With increasing size and/or quantity of solids, dispersion is coarser, yielding hetergeneous behaviour, i.e. with a pronounced solids concentration gradient along the vertical axis of the pipe. The actual velocity of flow then becomes a critical parameter.

With complex flow, some of the solids content behaves heterogeneously in pseudo-homogeneous flow, i.e. the flow can be described as homo-heterogeneous. These are thus two separate sources of friction and pressure drop.

Figure 3 shows the likely regimes for heterogeneous and homogeneous flow with typical slurries related to particle size and solids specific gravity for flow velocities in the range 1.2 to 2.6 m/sec (4 to 8 ft/sec).

Homogeneous flow

Common practice with slurries is to use the Fanning friction factor to estimate frictional losses. This is a quarter the value of the D'Arcy–Weisbuck friction factor. However, this straightforward approach does not take into account the fact that solids present have the effect of suppressing turbulence which can reduce the actual friction factor by up to 15%, depending on the type of slurry. Empirical formulae can thus be more realistic.

Pseudo-homogeneous flow

Pseudo-homogeneous flow is considered to exist where there is no measurable solids concentration gradient along the vertical axis of the pipe. For any given mixture this is related to the flow velocity. Below the critical velocity, flow will be heterogeneous; above the critical velocity, flow will be pseudo-homogeneous.

Specifically, the flow condition can be expressed in terms of the C/C_A ratio where C is the solids concentration measured at an arbitrary part near the top of the pipe (usually 8% of the pipe diameter), and C_A is the solids concentration at the centre of the pipe.

If these two values are equal (i.e. $C/C_A = 1$), flow will be homogeneous. Progressively lower values represent pseudo-homogeneous flow, degenerating into heterogeneous flow. The actual value of C/C_A is influenced by particle size and concentration, as well as flow velocity. In a mixture of solids, finer particles will have a high C/C_A and coarser particles a low C/C_A. As a general guideline, a C/C_A of 0.8 or greater is necessary to maintain pseudo-homogeneous flow.

Heterogeneous flow

With C/C_A values below 0.8 flow will be a mixture of pseudo-homogeneous and heterogeneous, e.g. fewer particles remaining in suspense with coarser particles tending to settle out. Flow will be fully heterogeneous at C/C_A values of 0.1 or less. With heterogeneous flow, inertia effects are far more significant than viscous effects. Also, there may be several different flow patterns ranging from symmetric suspension through asymmetric suspension to sliding bed (solids sliding along the bottom of the tube), then stationary bed, finally leading to plugging (pipe blockage).

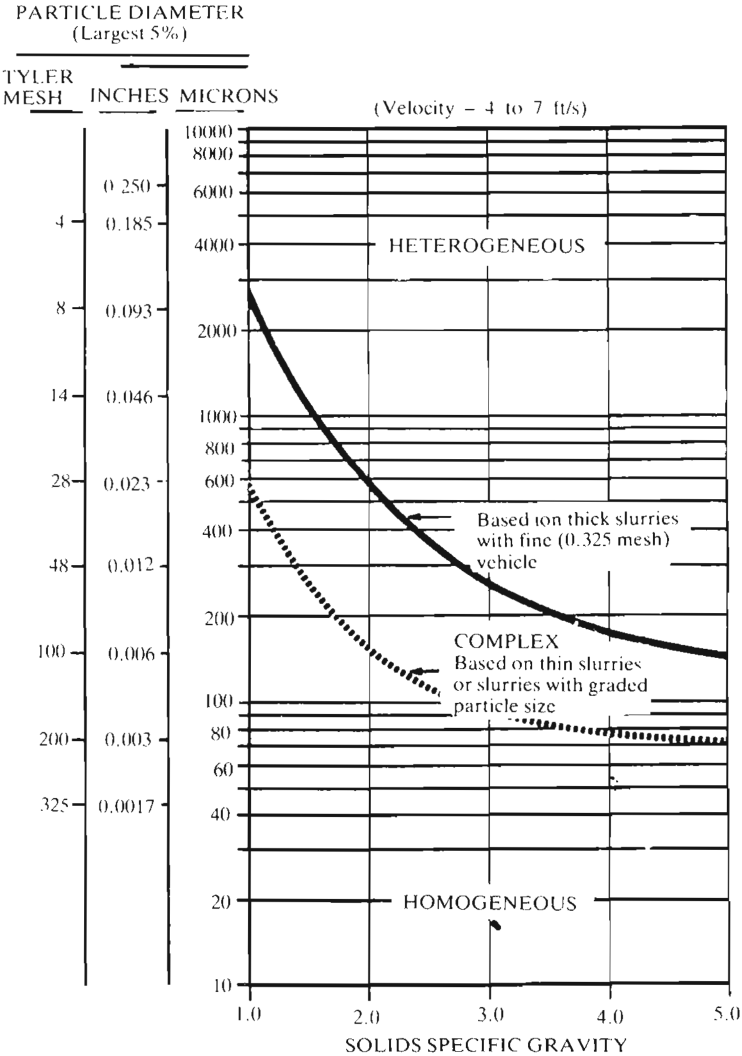


Figure 3. Regimes for homogeneous and heterogeneous flow.

Friction losses for heterogeneous flow are commonly based on the Durand formula, although empirical formulae are also used (see later). The Durand formula for the friction factor (f_h) for heterogeneous flow is:

$$f_h = f_l\left(1 + \frac{150Dg}{V^2}\right)\left(\frac{\rho S - \rho l}{\rho l}\right)\left(\frac{1}{\sqrt{C_D}}\right)^{3/2} \times S_V$$

where f_l = friction factor for liquid (dimensionless)
D = pipe diameter (ft)
V = flow velocity (ft/sec)
ρ_l = density of liquid (lb/ft^3)
ρS = density of solids (lb/ft^3)
C_D = drag coefficient (dimensionless)
S_V = volume friction of solids (dimensionless)
S_V = volume fraction of solids (dimensionless)

Homo-heterogeneous flow

With homo-heterogeneous flow, some of the solids behave heterogeneously in a homogeneous vehicle. This is a condition commonly encountered in practice where the carrier fluid contains a mixture of particle sizes. To determine friction losses in this case, it is necessary to split the solids content into fractions of different size and into homogeneous and heterogeneous portions. i.e. based on the respective C/C_A ratios.

Thus, taking each sign fraction in turn and determining its C/C_A ratio, multiplying this by volume concentration for that friction will give the proportion of that fraction having homogeneous flow. The remainder will be heterogeneous flow. Each fraction is split into homogeneous and heterogeneous flow in a similar manner. Friction losses are then calculated for each flow.

Transition velocity

Normally, all slurry pipeline systems operate with turbulent flow. Operating under laminar-flow conditions will allow some settlement which in time can lead to unstable flow conditions, or even blockage. Flow velocities must therefore be above the transition velocity, determined by the critical Reynolds number.

Transition velocities for Bingham plastic fluids are conveniently related to a dimensionless Hedstrom number (N_{HI}), where:

$$NHI = \text{Reynolds number} \times \text{plastic number}$$

$$= \left(\frac{\rho VD}{\eta}\right)\left(\frac{\frac{To g}{\eta}}{V/D}\right)$$

where ρ = slurry density (lb/ft^3)
V = flow velocity (ft/sec)
D = pipe diameter (ft)
T_O = yield spec (lb/ft^2)
g = acceleration of gravity
η = coefficient of rigidity (lb/ft sec)

The relationship between critical Reynolds number and Hedstrom number is shown in Figure 4 and is closely followed by most slurries. Mud and clay slurries can be the exception.

Critical deposition velocity

The critical deposition velocity relative to heterogeneous flow is given by Durand as:

$$V_{crit} = K\frac{2gD(\rho_s - \rho_l)^{1/2}}{\rho_l}$$

where K is an empirical constant
g = acceleration of gravity
D = pipe diameter (ft)
ρ_s = density of solids (lb/ft^3)
ρ_l = density of liquid (lb/ft^3)

In all such cases, performance calculations can be based on a pseudo-viscosity or equivalent Newtonian viscosity.

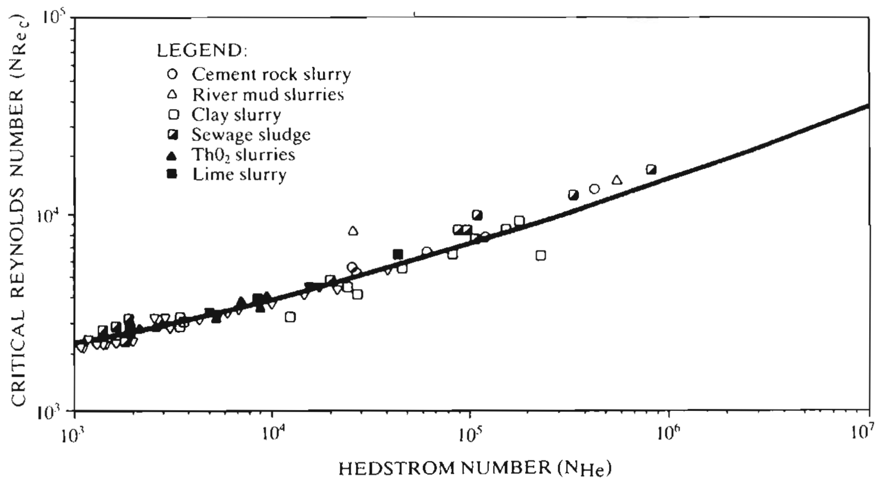


Figure 4. Variation of N_{Re_c} with N_{He} for Bingham flow in pipes.

Slurries

Slurries are liquids (usually water) containing abrasive solids in suspension, resulting in an increase in specific gravity over the carrier fluid. Slurries are categorised by the size of the solids as fines, sands and coarse.

Slurries behave as non-Newtonian fluids with an apparent viscosity depending on the degree of suspension, which in turn is dependent on the flow rate. This may be generally related to a fall velocity or the minimum flow rate necessary to maintain the solids in suspension and prevent them from settling out. This, in turn, depends on the size of the solids, and also their concentration. Approximate flow velocities to retain solids in suspension in water for various classes of slurries are:

Fines	(particle size 75 μm or less)	0.9 m/sec	(3ft/sec)
Sands	(particle size 75 to 850 μm)	1.5 m/sec	(5ft/sec)
Coarse	(particle size 850 to 5000 μm)	2.1 m/sec	(7ft/sec)

These empirical figures are based on a solids content of 30 to 35% by weight and solids of specific gravity 2.5 to 3.0 (see also Table 1 and Figure 5).

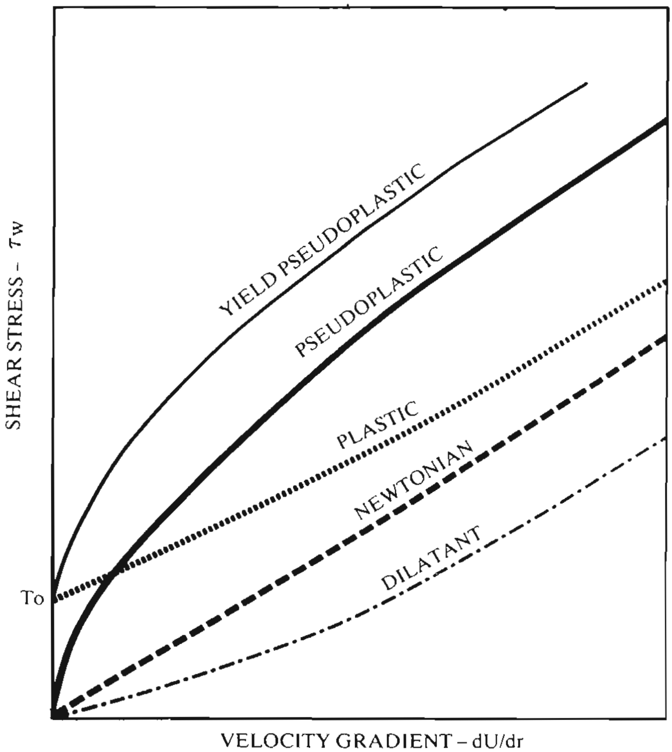


Figure 5. Fluid classification of slurries.

Table 1. Some typical slurries

Slurry	Proportion of solids by weight
Alumina	up to 50%
Crushed chalk	up to 68%
Clay	up to 60%
Coke fines	up to 55%
Gravel	up to 25%
Lime	up to 65%
Magnetite	up to 60%
Sand	up to 60%
Soda ash	up to 60%

Solids which form an intimate mixture yield a homogeneous fluid, in which case the pump performance is largely determined by the specific gravity and viscosity of the homogeneous mixture, which behaves as a normal fluid. Solids in suspension, however, form non-homogeneous mixtures and flow is then heterogeneous, with particles tending to slide at the surface of a 'bed' of solids. This 'bed' is only carried fully suspended if the fluid velocity is greater than the settling velocity of the solids involved. At lower fluid velocities there will be a corresponding degree of settlement producing a sliding rather than a suspended 'bed'.

The quantity of water required to deliver a specific quantity of solids when pumping slurries can be determined from:

$$\text{Water quantity} = K \times T\left(\frac{W}{R} + \frac{1}{sg_D}\right)$$

where T = weight of dry solids/hr
W = percentage of water
R = percentage of dry solids
For water quantity in litres and T in metric tonnes K = 1.06
For water quantity in Imperial gallons and T in Imperial tons K = 3.75
For water quantity in US gallons and T in US tons K = 4.02
sg_D = specific gravity of dry solids

Specific gravity figures for suspensions of solids in water are given in Table 2.

Frictional losses

There is no complete agreement on the method of calculating the frictional losses in pipes carrying fluids with solids in suspension. Generalised data can give extremely inconsistent results when applied to individual systems, particularly if localised areas exist where the fluid velocity may be less than the minimum needed to keep the solids in suspension. It can thus prove

Table 2. Specific gravity of suspensions of solids in water

Percentage by weight of solids	Ratio water to solids	Specific gravity of dry solids																	
		2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.5	4.0	4.5	5.0
		Specific gravity of solution																	
10	9:1	1.05	1.05	1.06	1.06	1.06	1.06	1.06	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.08	1.08	1.09	1.09
15	5.66:1	1.08	1.08	1.09	1.09	1.09	1.10	1.10	1.10	1.11	1.11	1.11	1.11	1.11	1.12	1.12	1.13	1.13	1.14
20	4:1	1.11	1.11	1.12	1.12	1.13	1.14	1.14	1.14	1.15	1.15	1.15	1.16	1.16	1.16	1.17	1.18	1.19	1.19
25	3:1	1.14	1.15	1.15	1.16	1.17	1.18	1.18	1.19	1.19	1.19	1.20	1.20	1.21	1.21	1.22	1.23	1.24	1.25
30	2.33:1	1.17	1.18	1.19	1.20	1.21	1.22	1.23	1.23	1.24	1.24	1.25	1.25	1.26	1.26	1.27	1.29	1.31	1.31
35	1.87:1	1.21	1.22	1.23	1.25	1.26	1.27	1.28	1.28	1.29	1.30	1.30	1.31	1.32	1.32	1.33	1.35	1.37	1.39
40	1.5:1	1.25	1.26	1.28	1.29	1.30	1.32	1.33	1.34	1.35	1.36	1.36	1.37	1.38	1.39	1.40	1.43	1.45	1.47
45	1.22:1	1.29	1.30	1.32	1.34	1.36	1.37	1.38	1.40	1.41	1.42	1.43	1.44	1.45	1.46	1.47	1.51	1.54	1.56
50	1:1	1.33	1.35	1.37	1.39	1.41	1.43	1.44	1.46	1.47	1.49	1.50	1.51	1.52	1.53	1.55	1.60	1.63	1.67
55	0.91:1	1.37	1.38	1.41	1.43	1.44	1.49	1.51	1.53	1.55	1.56	1.58	1.59	1.61	1.62	1.65	1.70	1.75	1.79
60	0.67:1	1.43	1.46	1.48	1.51	1.54	1.56	1.58	1.61	1.63	1.65	1.67	1.68	1.70	1.72	1.75	1.82	1.87	1.92
65	0.54:1	1.48	1.51	1.55	1.58	1.61	1.64	1.67	1.69	1.72	1.74	1.76	1.79	1.81	1.83	1.87	1.95	2.03	2.08
70	0.43:1	1.54	1.57	1.62	1.65	1.69	1.72	1.75	1.79	1.82	1.85	1.88	1.90	1.93	1.95	2.00	2.10	2.20	2.27
75	0.33:1	1.60	1.65	1.69	1.73	1.78	1.82	1.86	1.90	1.93	1.97	2.00	2.03	2.06	2.09	2.15	2.29	2.40	2.50

difficult to estimate for centrifugal pumps the total head to be supplied by the pump and thus to determine the most efficient working point.

A general formula which can be used is:

$$\frac{\Delta P_S}{\Delta P_W} - 1 = KC_V \qquad \frac{D(sg - 1) \times V_S}{\sqrt{d(sg - 1) \times V^2}}$$

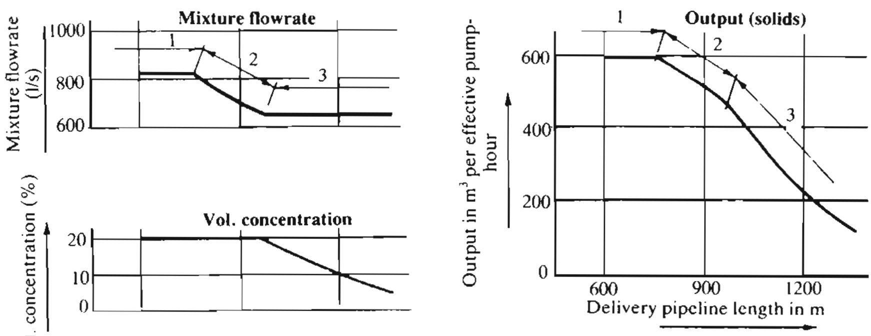
where the constant K is determined from emperical data

where ΔP_S = pressure drop when transporting solids
ΔP_W = pressure drop for water
C_V = concentration of solids by volume
D = pipe diameter
d = mean particle diameter
sg = specific gravity of solids
V_S = settling velocity of solids (in still water)
V = flow velocity

Output

Typically, the output from a slurry pump will follow the characteristics shown in Figure 6, with the output curve consisting of three zones:

(i) delivery distance,
(ii) delivery distance within the normal working range of the pump,
(iii) loss of output over any further delivery distance.



1. Delivery distance.
2. Delivery distance within the normal working range of the pump.
3. Loss of output over any further delivery distance.

Figure 6.

The first zone is critical in that it determines the suction conditions. At a high flowrate, when the upper limit is reached, the vacuum becomes critical, i.e. the flowrate at the attainable mean specific gravity of the mixture is so high that the corresponding vacuum on the suction side of the pump installation is equal to the critical vacuum, which may not be exceeded and thus constitutes the limit of the attainable vacuum. Figure 7 shows the critical vacuum applying to a delivery pipeline length L_1.

The critical vacuum is reached at working point P_1. If the critical vacuum is exceeded, e.g. as a result of shortening the delivery pipeline, with the result that a lower resistance curve applies (see L_2 in Figure 7), cavitation ensues.

If it is desired to operate with a shorter delivery pipeline, the speed n_1 of the pump must be reduced to the point where the intersection of the corresponding pump characteristic and that of the shorter pipleline falls within the normal working range of the pump. This implies that the point of intersection (working point P_3) must coincide with a flowrate at which the vacuum is slightly below the critical level corresponding to the lower pump speed. This is shown in Figure 7. The two working points P_1 and P_3 have a virtually equal flowrate. If the delivery distance is less and the specific gravity of the mixture remains virtually unchanged, the output at these (too) short delivery distances will also remain virtually constant, as shown by the horizontal section 1 of the output group (Figure 6).

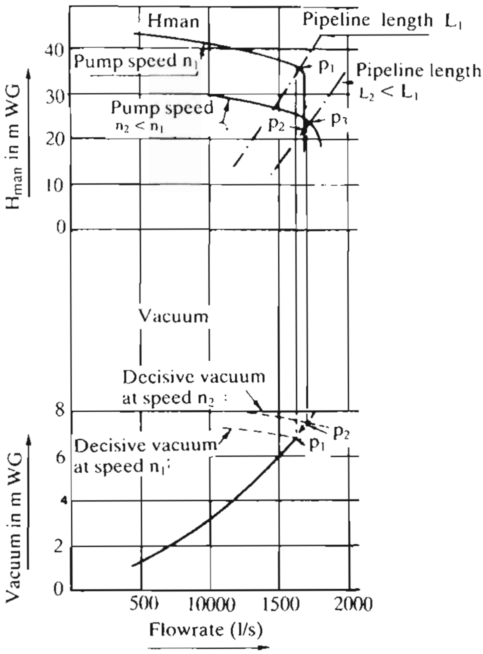


Figure 7.

Critical velocity

If the delivery pipeline is lengthened, the specific gravity of the mixture to be pumped must be reduced as soon as the critical velocity (the second bend in the output curve) is reached. This is necessary to avoid a subcritical situation, leading to sedimentation. At what is in effect an excessive delivery distance, pumping a mixture of such specific gravity as to cause sedimentation in the pipeline is to risk total blockage. In practice, this danger can be averted by admitting more water. The specific gravity must be reduced just sufficiently to restore a supercritical situation.

In approximate terms, it can be stated that the critical flowrate of a soil/water mixture at the reduced specific gravity differs little from the critical flowrate at the highest attainable mean specific gravity This implies that, when delivering into pipelines that are too long, the specific gravity must be reduced when a certain flowrate, which is virtually constant, is reached. This is the flowrate corresponding to the critical velocity. Increasing the delivery distance will therefore result in a lower output of solids.

When pumping mixtures of water and fine sand (less than 75 :m), or clay, e.g. soil and silt, or combinations of these materials, the critical velocity in the delivery pipe will be very low. Moreover, the resistance offered by the pipeline is less than would be the case during the transport of mixtures of water and coarser sand under comparable circumstances. As a result, the bend in the output curve, between line section 2 (the actual working range of the pump) and section 3 (the area relating to delivery distances that are too long), will coincide with an extremely high delivery-distance value, or will be missing altogether. The output curve will then be as shown in Figure 8 and consist of only two sections.

Sludge

Sludge is defined as a liquid (usually water) containing large solids with a particle size of 6 mm ($^1/_4$ in) or greater, the solids being soft rather than abrasive in nature. These solids may be further describes as 'stringy', 'clogging', etc., although a more useful classification would be 'soft' and 'hard' sludges because the solids may be hard and abrasive in some cases. Sludges may also contain a proportion of smaller solids or sand which could have an abrasive effect, affecting pump material choice, clearances, etc. Thus, sludges may have some of the characteristics of slurries. Sewage, on the other hand, mostly involves soft solids.

Frictional factor

Frictional losses involved in the transport of sludges and sewage are difficult to evaluate other than on empirical lines. However, where the mixture is reasonably homogeneous, a friction factor may be calculated on the basis of a pseudo-Reynolds number. This takes the form:

$$R_e = \frac{AQ}{C^x d^y}$$

where Q is the flow rate
A is an empirical factor
C is the consistency of the mixture
d is the solids' diameter

Values of the exponentials x and y are determined empirically for different fluids. Typical values for pulps are $x = 1.157$ and $y = 1.795$.

The friction factor, for insertion in standard function formulae, is then determined as:

$$fs = \frac{K}{(R_e{}^z)}$$

where K is a constant for a particular sludge.

The value of the exponential z also varies with the type of sludge, but is typically of the order of 1.63.

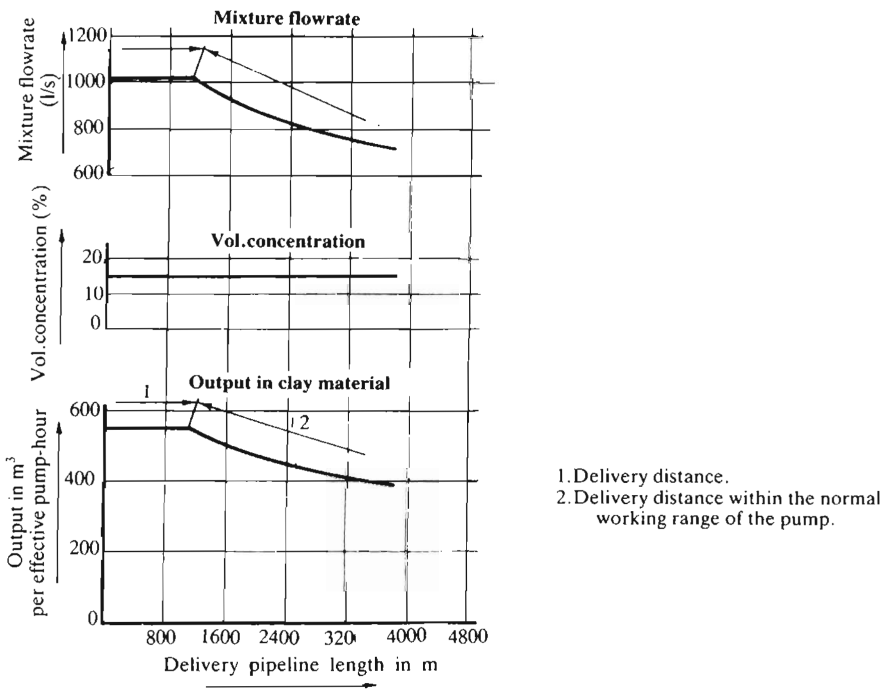


Figure 8.

Table 3. Head-capacity factors for stock

Stock consistency (%)	Head factor (H_F)	Capacity factor (Q_F)	Head-capacity factor (H_F/Q_F)
1.0	1.00	0.99	0.99
2.0	1.00	0.99	0.99
2.5	1.00	0.98	0.98
3.0	1.00	0.97	0.97
3.5	0.99	0.96	0.95
4.0	0.98	0.92	0.90
4.5	0.97	0.87	0.85
5.0	0.95	0.80	0.76
5.5	0.93	0.72	0.67
6.0	0.90	0.62	0.56
6.5	0.87	0.52	0.45
7.0	0.83	0.42	0.35

(a)

(b)

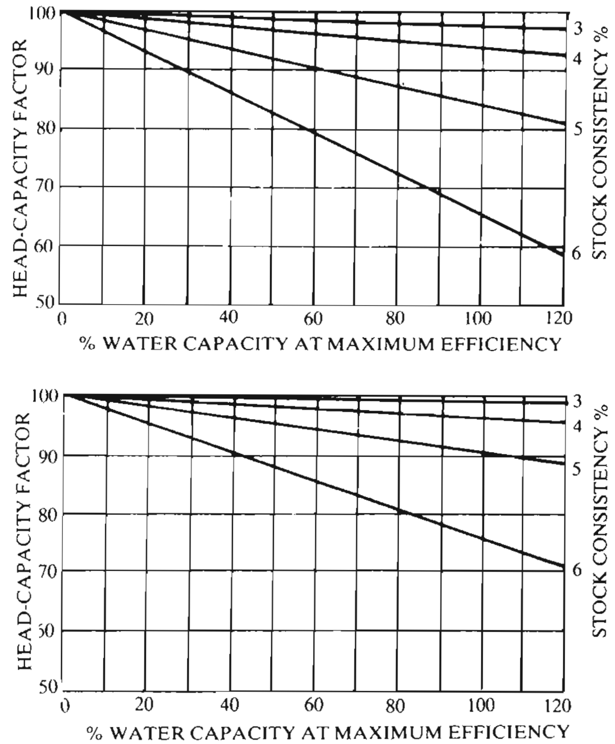


Figure 9. Determination of head-capacity correction factors: (a) chemical stock; (b) mechanical and reclaimed stock.

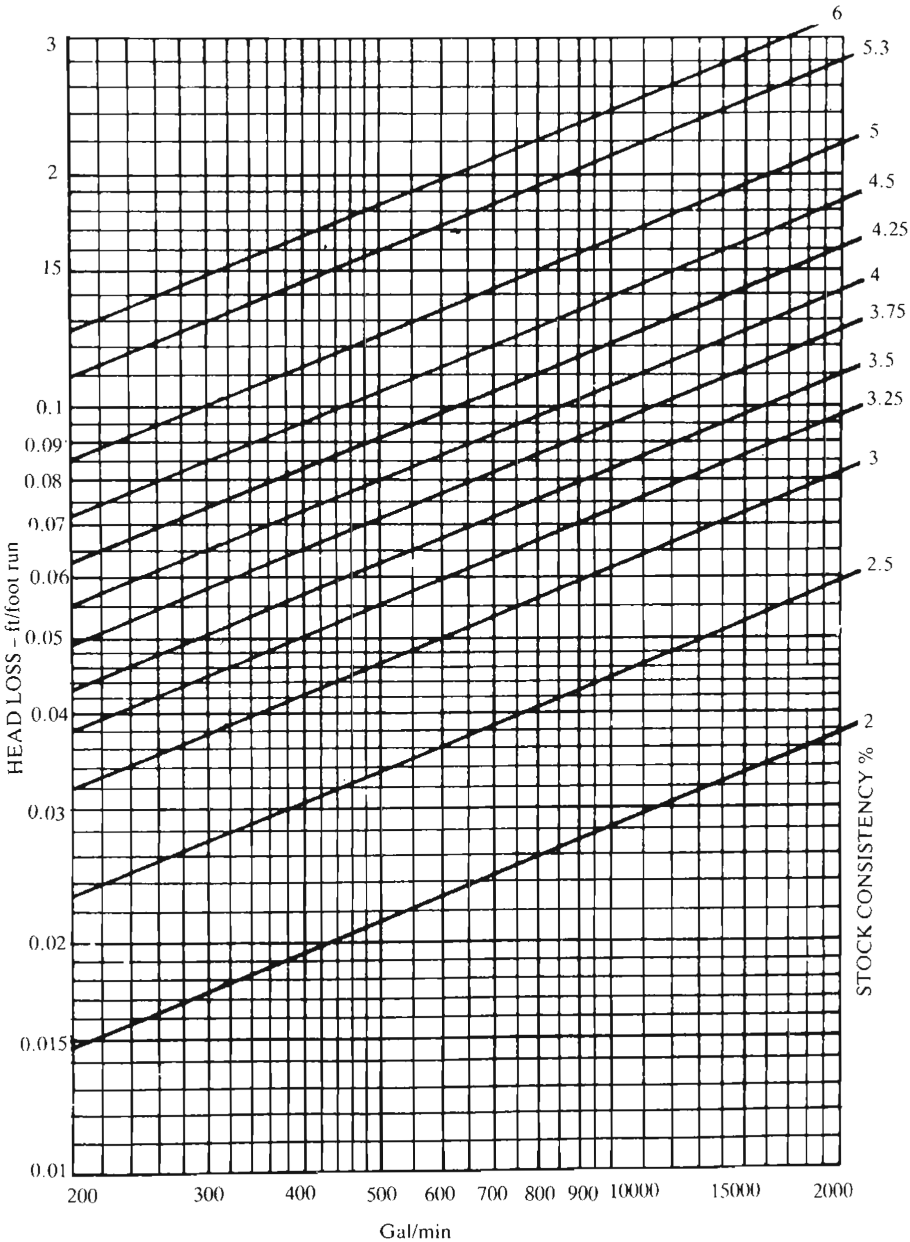


Figure 10. Friction loss with pulp stock: 12-in pipe.

Paper stock (pulp)

Paper stock is basically in the form of sludge with a specific type of solids (paper pulp). With low consistencies (i.e. less than 1% pulp by weight), flow and friction losses can be calculated as for water. With higher consistencies there is an increasing derating of pump performance (see Table 3).

Frictional losses increase rapidly with stock consistencies above about 2% by weight. A friction factor can be determined based on a pseudo-Reynolds number, viz:

$$f = \frac{K}{R_e{}^x} = \frac{K^1 Q}{C^y D^z}$$

where K = an empirical constant, dependent on the type of pulp
R_e = pseudo-Reynolds number
K^1 = a constant depending on the units employed
C = stock consistency %
D = pipe diameter
x, y and z are exponents with the typical values:
$x = 1.63$
$y = 1.16$
$z = 1.8$.

For Q in gal/min and D in ft
$K^1 = 17.2$

For Q in l/min and D in cm
$K^1 = 8.18$

(see also Figures 9 and 10).

Compressible Flow in Pipes

Air, steam and gases are all compressible fluids and the D'Arcy equation used for determining pressure and head losses with liquid flow is no longer applicable because the density of gases and vapours changes considerably with changes of pressure. However, for simplified general engineering calculations not requiring great accuracy, liquid D'Arcy flow formulae may be used if the pressure drop involved is less than 10% of the inlet pressure. Use of such formulae is also sometimes extended to pressure drops up to 40% of the inlet pressure, provided in this case the specific volume is taken as the average of the upstream and downstream conditions.

The real flow of a pressurised gas through pipes differs appreciably in a number of important characteristics from the flow of liquids in pipes. Pressure, for example, drops at an increasing rate along the pipe, rather than with a constant pressure gradient. At the same time, velocity tends to increase up to a maximum defined by V = %kgRT for air, but subject to a limiting or maximum length, which must correspond to the end of the pipe. At this point the pressure gradient in infinite, i.e. the pipe is effectively closed. In this equation, k is the ratio of specific heats at constant pressure to constant volume, R the individual gas constant, and T the absolute temperature (degrees Rankine). An alternative formula is M = 1 %k (=0.845 for air), where M is the Mach number.

The general equation may be written in the same form as the D'Arcy equation for fluid flow, but with the addition of an extra term representing the pressure drop required to increase the flow momentum:

$$\frac{\Delta P}{L} = \frac{f}{D} \times \frac{\rho V^2}{2} + \beta\rho\bar{Q}\frac{dV}{dL}$$

where ΔP is the pressure drop
- L is the pipe length
- f is a constant friction (but dependent on surface roughness)
- D is the pipe diameter
- ρ is the gas density
- $\bar{Q}$ is the specific volume of gas
- dV/dL is the velocity gradient

β is a factor of the order of unity and normally taken as 1.0 (i.e. can be eliminated from the equation).

Actual flow conditions may range from adiabatic to isothermal. Adiabatic conditions are only likely to apply in short, well-insulated pipes where no appreciable heat is transferred to or from the pipe. Isothermal flow, or flow at constant temperature, is commonly assumed as more consistent with normal practice, especially for long pipes. In fact, most practical pipelines will generate polytropic flow conditions, which are virtually impossible to analyse. The assumption of isothermal flow is thus a practical compromise.

Isothermal flow

With isothermal flow, a formula developed from basic principles is:

$$\frac{\Delta P}{L} = \frac{\frac{f}{D} \times \frac{\rho V^2}{2}}{kM^2 - 1}$$

The friction factor is dependent on the Reynolds number of flow and pipe roughness. It can be assumed independent of Mach number.

The Reynolds number will be constant for isothermal flow, but may vary with adiabatic or isentropic flow (and certainly with diabatic flow), in which case a mean value can be assumed.

The Reynolds number is a dimensionless quantity, given arithmetically by:

$$\begin{aligned} R_e &= \frac{\rho VD}{\mu} \\ &= \frac{VD}{\nu} \end{aligned}$$

$$\begin{aligned} \text{where } \rho &= \text{the mass density} \\ \mu &= \text{viscosity} \\ \nu &= \text{kinematic viscosity} \end{aligned} \left\{ \begin{array}{l} \text{in units consistent with} \\ \text{velocity (V) and} \\ \text{pipe diameter (D)} \end{array} \right.$$

For air at standard temperature $\nu = 1.568 \times 10^{-4}$ ft^2/sec.

$$\begin{aligned} \text{Thus } R_e &= 531.5 \text{ VD} \\ &= 530 \text{ VD (with sufficient practical accuracy).} \end{aligned}$$

where V is in ft/sec
D is in inches.

The friction factor (f) is common for all fluids (i.e. gases and liquids) and is normally determined from empirical charts. For laminar flow, the friction factor is dependent only on Reynolds number and is numerically equal to $64/R_e$, laminar flow being defined by the Reynolds number not exceeding 2000. For turbulent flow and smooth-bore pipes, the Reynolds number can be calculated from the following empirical formula:

$$f = \frac{0.3164}{R_e^{0.25}}$$

As the Mach number (M) approaches 0, the denominator in this equation comes closer and closer to unity, reducing the equation to the same as that for liquid flow. There is some justification for using the D'Arcy formula for compressible-flow calculations (as mentioned initially) as M60 (i.e. consistent with short lengths of pipes with resulting low pressure drops). This also implies that at very low Mach numbers, compressible flow can be treated as incompressible. Pressure gradients ($\Delta P/L$) will, in fact, be within 5% of incompressible flow values at Mach numbers up to about 0.18 for air.

A complete working formula for isothermal flow, expressed in terms of mass flow rate (Qm) is:

$$Qm^2 = \left[\frac{144gA^2}{\left(\bar{V}_1 \frac{fL}{D} + 2\log\frac{P_1}{P_2}\right)}\right]\left[\frac{P_1^2 - P_2^2}{P_1}\right]$$

where A = cross-sectional area of pipe (ft^2)
g = acceleration of gravity = 32.2 (ft/sec^2)
$\bar{V}_1$ = specific volume of gas (ft^3/lb)
L = length of pipe (ft)
f = friction factor
D = pipe bore (ft)
P_1 = absolute pressure (entry)
P_2 = absolute pressure (exit)

For long gas pipelines (where velocity gradient can be ignored) this reduces to:

$$Qm^2 = \left(\frac{144gDA^2}{\bar{V}_1 f \times L}\right) \times \left(\frac{P_1^2 - P_2^2}{P_1}\right)$$

Or expressed in terms of *volume flow rate* (Q_V) in ft^3/sec:

$$Q_V = 114.2\sqrt{\left(\frac{P_1^2 - P_2^2}{fLT \times sg} \times D^5\right)}$$

where T is the absolute temperature (degrees Rankine)
sg is the specific gravity of the gas.

Other working formulae of similar derivation are:

Weymouth formula:

$$Q_V = 28.0D^{2.667}\sqrt{\left(\frac{P_1^2 - P_2^2}{L\,sg}\right)\frac{520}{T}}$$

Panhandle formula:

$$Q_V = 36.8ED^{2.6182}\sqrt{\left(\frac{P_1^2 - P_2^2}{L}\right)^{0.5394}}$$

where E is an efficiency factor, normally taken as 0.92 for average conditions.

The Panhandle formula is widely used for natural gas pipelines from 6 to 24 in diameter.

Limiting values

Maximum possible velocity (V^*) in a pipe is source velocity ($M = 1$), given directly by:

$$V^* = 12\sqrt{kg\,PV_V}$$

where V^* is in ft/sec
k is the ratio of specific heats of the gas
P is the absolute pressure (lbf/in^2)
V_V is the specific volume of gas (ft^3/lb)

Maximum pressure (P^*) can be determined on the basis of continuity, viz:

$$V_1P_1 = VP = V^*P^*$$

Hence

$$\frac{P^*}{P_1} = \frac{V_1}{V^*} = \frac{M_1}{M^*} = M_1\sqrt{k}$$

where the suffix 1 refers to initial concentration.

Hence (and because the velocity of sound is constant under isothermal conditions):

$$P^* = P_1 \times M_1k$$

Unity Length (L^*) can be determined from the general equation:

$$\frac{fL^*}{D} = \left(\frac{1}{kM_1^2} - 1\right) - \log_e \frac{1}{kM_1^2}$$

or

$$L^* = \frac{D}{f}\left(\left(\frac{1}{kM_1^2} - 1\right) - \log_e \frac{1}{kM_1^2}\right)$$

The implication of this is that velocity of gas flow can go on increasing in a pipe up to a maximum Mach number of $1/\sqrt{k}$. This increase ceases at a limiting length of pipe (L^*), which must be at the end of the pipe. If the actual length of pipe is greater than L^*, initial conditions will have to be adjusted to reduce M_1 so that the actual pipe length is less than, or equal to, L^*.

At the same time, the limiting pressure ratio (P^*/P_1) and limiting length (L^*) are dependent only on initial velocity (Mach number) and the k value of the gas.

Pressures and lengths between two sections of a pipeline can thus be expressed as follows (if the Mach numbers at each section are known):

$$\frac{P_2}{P_1} = \frac{M_1}{M_2}$$

$$\frac{fL}{D} = \left(\frac{fL^*}{D}\right)M_1 - \left(\frac{fL^*}{D}\right)M_2$$

Adiabatic flow

Similar treatment applies for adiabatic flow, although the corresponding formulae are more complicated and may require working as a series of approximations in order to reach real values in particular cases. In general, the pressure, temperature and velocity will always be slightly less than those for isothermal flow, but the limiting length will be similar. The differences are usually small enough to be negligible, except at higher Mach numbers, and thus, for simplification of calculations, isothermal formulae can be used for subsonic adiabatic flow.

Limiting velocity:

This is given directly by:

$$\frac{V^*}{V_1} = \frac{1}{M_1}\sqrt{\frac{2\left(1 + \frac{k-1}{2}M_1^2\right)}{k+1}}$$

Limiting temperature:

$$\frac{T^*}{T_1} = \frac{2\left(1 + \frac{k-1}{k2} M_1^2\right)}{k+1}$$

Limiting length:

$$L^* = \frac{D}{F}\left(\frac{1 - M_1^2}{kM_1^2}\right) + \frac{k+1}{2k} \log_e \left[\frac{(k+1)M_1^2}{2\left(1 + \frac{k-1}{2} M_1^2\right)}\right]$$

Limiting pressure:

$$\frac{P^*}{P_1} = M_1 \left[\frac{2\left(1 + \frac{k-1}{2} M_1^2\right)}{k+1}\right]$$

Stagnation state

Flow is possible between two extremes. At one extreme, velocity is zero and temperature is a maximum because all the kinetic energy is converted to enthalpy. The speed of sound is also a maximum (stagnation point or stagnation state). At the other extreme, the velocity is a maximum and the temperature falls to absolute zero, all the enthalpy being converted into kinetic energy. The speed of sound is then zero (zero temperature state). Between these extremes the practical flow may be subsonic, transonic or supersonic (Figure 1) although the zero temperature state can never be reached (i.e. it is a hypothetical condition).

At the stagnation state:

$$\frac{0^2}{2} + h_0 = \frac{V^2}{2} + h$$

or

$$h_0 = \frac{V^2}{2} + h$$

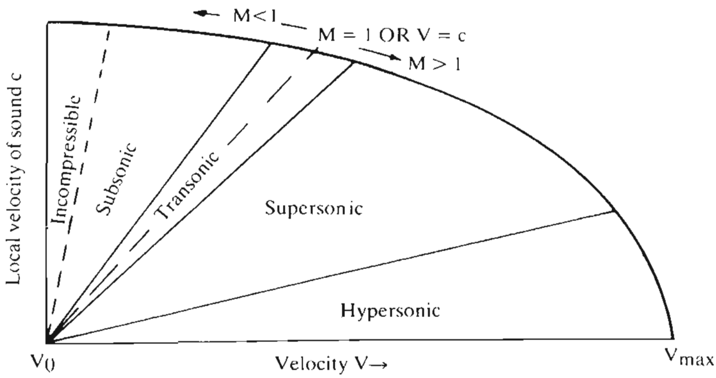


Figure 1.

From the general gas relationship it follows that the stagnation temperature (T_O) is given by:

$$T_O = T + \frac{V^2}{2sp}$$

where sp = specific heat at constant pressure.

Alternatively,

$$\frac{T_O}{T} = 1 + \frac{k-1}{2} \times M^2$$

where M = Mach number $= \frac{V}{c}$.

The stagnation pressure can be derived as:

$$\frac{P_O}{P} = \left(\frac{T_O}{T}\right)^{\frac{k}{k-1}} = \left(1 + \frac{k-1}{2} M^2\right)^{\frac{k}{k-1}}$$

This may be expanded in the form:

$$P_O = P + \frac{\rho V^2}{2}\left\{1 + \frac{M^2}{4} + \frac{2-k}{24} M^4 + \ldots\right\}$$

This can be compared with the equation for incompressible flow:

$$P_O = P + \frac{\rho V^2}{2}$$

The difference between these two equations represents the effect of increased gas density due to compressibility, generally termed the compressibility factor. Values of the compressibility factor range from unity at very low Mach numbers (where there are no compressibility effects), up to 1.276 as the Mach number approaches 1 (velocity approaches the speed of sound in the gas). Besides increasing the dynamic pressure of compressible flow, compared with incompressible flow, the rising value of the compressibility factor can also affect the flow velocity through ducts with varying area. The relationship between area and velocity changes is, in fact, a function of the local Mach number, and can be rendered in the form:

$$\frac{dA}{dV} = \frac{A}{V}(M^2 - 1)$$

With subsonic flow, a decrease in area produces an increase in flow velocity and vice versa (similar to incompressible flow), i.e. area and Mach number changes are opposite. The flow velocity may be sonic only at a constant section. With supersonic flow, a decrease in flow area produces a decrease in flow velocity and vice versa, i.e. area and Mach number changes are the same.

Flow from stagnation conditions

Gas compressed and stored in a reservoir is essentially under stagnation conditions, where velocity is zero and the pressure and temperature are known (or can be determined). Where the reservoir is used as a supply, the velocity, temperature and pressure at any other section of flow are determined basically from the following relationships:

Velocity at any arbitrary section:

$$V = \sqrt{2sp\,T_O\left(1 - \frac{P}{P_O}\right)^{\frac{k}{k-1}}}$$

Alternatively, for adiabatic flow, the velocity at any section can be determined from the temperature at that section:

$$V = \sqrt{2sp\,T_O\left(1 - \frac{T}{T_O}\right)}$$

Flow rate

Flow rate can be determined as the mass flow, i.e. mass flow = $VA\rho$, or directly as the product of V and A in numerically consistent units:

$$\text{Dimensions are } \frac{L}{T} \times L^2 = \text{flow rate} = \frac{L^3}{T}$$

Pressure at any arbitrary section:

$$P = \frac{P_O}{\left(1 + \frac{k-1}{2} \times M^2\right)^{\frac{k-1}{k}}}$$

Temperature at any section:

$$T = \frac{T_O}{1 + \frac{k-1}{2} \times M^2}$$

At any (constant) section where the flow is sonic, the flow conditions are described as critical, yielding a critical temperature (T^*) and a critical pressure (P^*), where:

$$\frac{T^*}{T_O} = \frac{2}{k+1} \quad \text{(adiabatic or isentropic flow)}$$

$$\frac{P^*}{P_O} = \left(\frac{2}{k+1}\right)^{\frac{k}{k-1}} \quad \text{(isentropic flow only)}$$

Note: For air, where $k = 1.4$, the value of critical pressure is $\left(\frac{2}{2.4}\right)^{\frac{1.4}{0.4}} = 0.528$.

That is, the critical pressure is 52.8% of P_O. Similarly, the critical temperature can be calculated as 83.3% of T_O.

Critical area:

The relationship between the critical area (A^*) or throat area where $\mu = 1$ and the area of any other section (A) is given by:

$$\frac{A}{A^*} = \frac{1}{M} \frac{\left(1 + \frac{k-1}{2} \times M^2\right)^{\frac{k+1}{2(k-1)}}}{\frac{k+1}{2}}$$

$$= \frac{1}{M}\left(\frac{1 + 0.2M^2}{1.2}\right)^3 \quad \text{for air.}$$

Nozzle flow

Flow at the throat of a nozzle, supplied by a reservoir or similar source under stagnation conditions, will be sonic if the critical pressure is greater than the receiver pressure (Figure 2a). This means that the flow will be critical. The flow velocity follows from calculating the critical temperature, from which:

Flow velocity $c = 49\sqrt{T^*}$ ft/sec

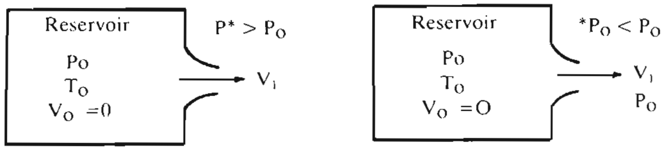


Figure 2.

(The velocity of sound in air = c = 49T ft/sec where T is the absolute temperature in degrees Rankine.)

If the critical pressure is less than the receiver pressure, then the flow cannot be critical (Figure 2b). In this case the flow will be subsonic and the exit pressure will equal the receiver pressure. The temperature can be calculated from the general formula, or from:

$$T_1 = T_O\left(\frac{P_1}{P_O}\right)^{\frac{k-1}{k}}$$

The velocity is likewise calculated from the general formula.

Similar analysis applies where the nozzle is of convergent–divergent form (Figure 3). In this case it is necessary to establish whether the flow is critical or not (at the throat). The throat velocity can be determined accordingly, and from this the final exit velocity from the divergent section.

Flow through a nozzle can also be rendered directly in terms of flow rate and a discharge coefficient, this being a convention for engineering calculations. The complete nozzle formula is:

$$\text{Mass flow} = A \times Ec \times \delta \times d^2 \sqrt{h}\,\sqrt{P_2/T}$$

where A = a constant depending on the units employed

E = coefficient for the velocity of approach

$$= \frac{1}{\sqrt{1 - m^2}}$$

$$\text{where } m = \frac{\text{cross-sectional area of nozzle}}{\text{cross-sectional area upstream}} = \frac{A_2}{A_1}$$

c = nozzle coefficient

δ = expansibility factor allowing for the change in air density which occurs during acceleration through the nozzle

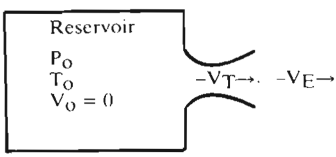


Figure 3.

$$= \frac{1 - 0.07h}{13.6P_2} \text{ for values of circa } 0.16$$

(and h in inches wg and P_2 in inches of mercury)

d = diameter of nozzle
h = pressure drop across nozzle
P_2 = absolute pressure on downstream side of nozzle
T = absolute temperature on downstream side of nozzle.

If T is in °R, P_2 in inches of mercury, h is in inches wg and d is in inches, a value of A = 0.1148 gives the mass flow in units of lb/sec.

For a specific nozzle profile, the formula can be simplified by the use of a nozzle constant appropriate to that particularly geometry and nozzle size. Rendered as a solution for conventional flow rate (Q):

$$Q = K(T_1/P_1)\sqrt{h}\,\sqrt{P_2/T_2}$$

where K = nozzle constant
T_1 = absolute temperature at specified inlet point
T_2 = absolute temperature at nozzle or specified point downstream
P_1 = absolute pressure at specified inlet point
P_2 = absolute pressure at nozzle or specified point downstream
h = pressure drop across nozzle.

Simplified orifice formulae

An orifice is a simple form of nozzle, formed by a circular hole cut in a thin flat plate. Flow can again be determined with reference to an empirical discharge coefficient or orifice coefficient. This will be much lower than for nozzles because of the less streamlined flow but, owing to the simpler form of the nozzle, will be less subject to variation. Thus nozzle coefficients may vary between 0.90 (or less) and 0.995, depending on size and geometry, whereas an orifice coefficient can be expected to be of the order of 0.61, regardless of size, and differing only if the orifice has a well-rounded, as opposed to a sharp, entry.

Very much simplified formulae can therefore be applied to assess the discharge of air through orifices, and the following are generally satisfactory for straightforward engineering calculations:

(1) For upstream pressures above 14.7 lbf/in^2g

$$Q\ (ft^3/min) \text{ for sharp-edged orifice} = \frac{218 \times A \times P_u}{\sqrt{460 + T}} \qquad (a)$$

or

$$= \frac{172 \times d^2 \times P_u}{\sqrt{460 + T}}$$

$$Q\ (\text{ft}^3/\text{min})\ \text{for rounded-entrance orifice} \simeq \frac{417 \times A \times P_u}{\sqrt{460+T}} \qquad \text{(b)}$$

or

$$\simeq \frac{327 \times d^2 \times P_u}{\sqrt{460+T}}$$

(2) For upstream pressures below 14.7 lbf/in²g

$$Q\ (\text{ft}^3/\text{min})\ \text{for sharp-edged orifice} = \frac{210 \times A \times P_u}{\sqrt{460+T}} \qquad \text{(c)}$$

or

$$= \frac{166 \times d^2 \times P_u}{\sqrt{460+T}}$$

$$Q\ (\text{ft}^3/\text{min})\ \text{for rounded-entrance orifice} \simeq \frac{324 \times A \times P_u}{\sqrt{460+T}} \qquad \text{(d)}$$

or

$$\simeq \frac{255 \times d^2 \times P_u}{\sqrt{460+T}}$$

where A = orifice area (in²)
P_u = upstream pressure (lbf/in²g)
d = orifice diameter (in)
T = upstream air temperature (°F)

1. (a) $Q\ (\text{ft}^3/\text{min}) = 11.9\, A \times P_u$
or $= 9.4\, d^2 P_u$
(b) $Q\ (\text{ft}^3/\text{min}) \simeq 18.3\, A \times P_u$
or $\simeq 14.4\, d^2 P_u$

2. (c) $Q\ (\text{ft}^3/\text{min}) = 11.5\, A \times P_u$
or $= 9.05\, d^2 P_u$
(d) $Q\ (\text{ft}^3/\text{min}) \simeq 17.7\, A \times P_u$
or $\simeq 13.92\, d^2 P_u$

Losses in Bends and Fittings

Pressure losses in a piping system due to changes in the shape of the flow path or changes in cross-section as produced by bends, valves, fittings, etc., can be evaluated in three different ways:

(i) as a resistance coefficient (K) for the component involved
(ii) as an equivalent length (L/D)
(iii) as a flow coefficient (C_v).

Resistance coefficients

Specifically because a bend, valve or fitting, etc., presents additional resistance to flow, there is a velocity head loss at that point which can be expressed directly as:

$$h_L = K\frac{V^2}{2g}$$

where H_L is the velocity head loss
K is the resistance coefficient for the component involved.

Table 1 gives a range of worked out values.

Pressure drop (ΔP) can also be calculated directly from resistance coefficient:

$$\Delta P = \frac{KV^2 w}{2g}$$

where w is the specific weight of the fluid.

For clean water: ΔP (lbf/in^2) $= 0.00673\ KV^2$
when v is in ft/sec
ΔP (bar) $= 0.000044\ KV^2$
where V is in m/sec

Table 1. Head loss in feet from resistance coefficient*

Resistance coefficient	Flow velocity—ft/sec (m/sec)									
	1 (0.3)	2 (0.6)	3 (0.9)	4 (1.2)	5 (1.5)	6 (1.8)	7 (2.1)	8 (2.4)	9 (2.75)	10 (3.0)
0.05	0.00080	0.003	0.007	0.013	0.019	0.030	0.039	0.050	0.063	0.077
0.1	0.00200	0.006	0.014	0.025	0.039	0.060	0.076	0.100	0.126	0.155
0.2	0.00300	0.012	0.028	0.050	0.078	0.112	0.152	0.199	0.252	0.310
0.3	0.00500	0.019	0.042	0.075	0.117	0.168	0.228	0.299	0.377	0.470
0.4	0.00600	0.025	0.056	0.100	0.155	0.224	0.304	0.398	0.503	0.620
0.5	0.00800	0.031	0.070	0.125	0.194	0.280	0.381	0.497	0.629	0.780
0.6	0.00900	0.037	0.084	0.149	0.233	0.335	0.457	0.597	0.755	0.930
0.7	0.01100	0.044	0.098	0.174	0.271	0.391	0.533	0.696	0.880	1.090
0.8	0.01200	0.050	0.112	0.199	0.311	0.447	0.609	0.796	1.006	1.240
0.9	0.01400	0.056	0.125	0.214	0.349	0.503	0.685	0.895	1.132	1.400
1.0	0.01600	0.062	0.140	0.249	0.388	0.559	0.761	0.995	1.258	1.550
1.1	0.01710	0.068	0.154	0.274	0.427	0.615	0.837	1.090	1.380	1.710
1.2	0.01860	0.074	0.168	0.299	0.466	0.671	0.913	1.190	1.510	1.860
1.3	0.02020	0.081	0.182	0.324	0.505	0.727	0.989	1.290	1.630	2.020
1.4	0.02170	0.087	0.196	0.349	0.544	0.783	1.065	1.390	1.760	2.170
1.5	0.02330	0.093	0.210	0.374	0.583	0.839	1.141	1.490	1.890	2.330
1.6	0.02480	0.099	0.224	0.398	0.621	0.895	1.217	1.590	2.020	2.480
1.7	0.02640	0.106	0.238	0.423	0.660	0.951	1.293	1.690	2.140	2.640
1.8	0.02800	0.112	0.252	0.448	0.699	1.007	1.369	1.790	2.270	2.800
1.9	0.02950	0.118	0.266	0.473	0.738	1.063	1.445	1.890	2.390	2.950
2.0	0.03106	0.124	0.280	0.497	0.776	1.118	1.522	1.990	2.520	3.110
2.1	0.03261	0.130	0.294	0.522	0.815	1.174	1.598	2.090	2.650	3.260
2.2	0.03416	0.136	0.308	0.547	0.854	1.230	1.674	2.190	2.780	3.420
2.3	0.03571	0.143	0.322	0.572	0.893	1.286	1.750	2.290	2.900	3.570
2.4	0.03726	0.149	0.336	0.597	0.932	1.341	1.826	2.390	3.030	3.730
2.5	0.03881	0.155	0.350	0.621	0.971	1.397	1.902	2.490	3.150	3.880
2.6	0.04036	0.161	0.364	0.646	1.009	1.453	1.978	2.590	3.280	4.040
2.7	0.04191	0.168	0.378	0.671	1.048	1.509	2.054	2.690	3.400	4.190
2.8	0.04346	0.174	0.392	0.696	1.087	1.565	2.130	2.790	3.530	4.350
2.9	0.04501	0.180	0.406	0.721	1.126	1.621	2.206	2.890	3.650	4.500
3.0	0.04659	0.186	0.419	0.746	1.165	1.677	2.283	2.990	3.770	4.660
3.1	0.04814	0.192	0.433	0.771	1.204	1.733	2.359	3.090	3.900	4.810
3.2	0.04969	0.198	0.447	0.796	1.243	1.789	2.435	3.190	4.030	4.970
3.3	0.05124	0.205	0.461	0.821	1.282	1.845	2.511	3.290	4.150	5.120
3.4	0.05279	0.211	0.475	0.845	1.321	1.901	2.587	3.390	4.280	5.280
3.5	0.05434	0.217	0.489	0.870	1.360	1.957	2.663	3.490	4.400	5.430
3.6	0.05589	0.223	0.503	0.895	1.398	2.013	2.739	3.590	4.530	5.590
3.7	0.05744	0.230	0.517	0.920	1.437	2.069	2.815	3.680	4.650	5.740
3.8	0.05899	0.236	0.531	0.945	1.476	2.125	2.891	3.780	4.780	5.900
3.9	0.06054	0.242	0.545	0.970	1.514	2.181	2.967	3.880	4.900	6.050
4.0	0.06212	0.248	0.559	0.994	1.553	2.236	3.044	3.980	5.030	6.210
4.1	0.06367	0.254	0.573	1.019	1.592	2.292	3.120	4.080	5.160	6.370
4.2	0.06522	0.260	0.587	1.044	1.631	2.348	3.196	4.180	5.290	6.520
4.3	0.06677	0.267	0.601	1.069	1.670	2.404	3.272	4.280	5.410	6.680
4.4	0.06832	0.273	0.615	1.093	1.709	2.460	3.348	4.380	5.540	6.830
4.5	0.06987	0.279	0.629	1.118	1.748	2.516	3.424	4.480	5.660	6.990
4.6	0.07142	0.285	0.643	1.143	1.786	2.572	3.500	4.580	5.790	7.140
4.7	0.07297	0.292	0.657	1.168	1.825	2.628	3.576	4.680	5.910	7.300
4.8	0.07452	0.298	0.671	1.193	1.864	2.684	3.652	4.770	6.040	7.450
4.9	0.07607	0.304	0.685	1.218	1.903	2.740	3.728	4.870	6.160	7.600
5.0	0.07765	0.311	0.698	1.243	1.942	2.795	3.806	4.97	6.29	7.77
5.1	0.07920	0.317	0.712	1.268	1.981	2.851	3.882	5.07	6.41	7.92
5.2	0.08075	0.323	0.726	1.293	2.020	2.907	3.958	5.17	6.54	8.08
5.3	0.08230	0.330	0.740	1.318	2.058	2.963	4.034	5.27	6.66	8.23
5.4	0.08388	0.336	0.755	1.342	2.097	3.019	4.110	5.37	6.79	8.39
5.5	0.08543	0.342	0.769	1.367	2.136	3.075	4.186	5.47	6.91	8.54

Table 1 (*continued*)

Resistance coefficient	Flow velocity—ft/sec (m/sec)									
	1 (0.3)	2 (0.6)	3 (0.9)	4 (1.2)	5 (1.5)	6 (1.8)	7 (2.1)	8 (2.4)	9 (2.75)	10 (3.0)
5.6	0.08698	0.348	0.783	1.392	2.175	3.131	4.262	5.57	7.04	8.70
5.7	0.08853	0.355	0.797	1.417	2.213	3.187	4.338	5.67	7.17	8.85
5.8	0.09008	0.361	0.811	1.442	2.252	3.243	4.414	5.77	7.30	9.01
5.9	0.09163	0.367	0.825	1.467	2.290	3.299	4.490	5.87	7.42	9.16
6.0	0.09318	0.373	0.839	1.491	2.329	3.354	4.567	5.97	7.55	9.32
6.1	0.09473	0.379	0.853	1.516	2.368	3.410	4.643	6.07	7.68	9.47
6.2	0.09628	0.385	0.867	1.541	2.407	3.466	4.719	6.17	7.81	9.63
6.3	0.09783	0.392	0.881	1.566	2.446	3.522	4.795	6.27	7.93	9.78
6.4	0.09938	0.398	0.895	1.590	2.485	3.572	4.871	6.37	8.06	9.94
6.5	0.10093	0.404	0.909	1.615	2.524	3.634	4.947	6.47	8.18	10.09
6.6	0.10248	0.410	0.923	1.640	2.562	3.690	5.023	6.57	8.31	10.25
6.7	0.10403	0.417	0.937	1.665	2.601	3.746	5.099	6.67	8.43	10.40
6.8	0.10558	0.423	0.951	1.690	2.640	3.802	5.175	6.76	8.56	10.56
6.9	0.10713	0.429	0.965	1.715	2.679	3.858	5.251	6.86	8.68	10.71
7.0	0.10871	0.435	0.979	1.740	2.717	3.913	5.328	6.96	8.80	10.87
7.1	0.11026	0.441	0.993	1.765	2.756	3.969	5.404	7.06	8.93	11.03
7.2	0.11181	0.447	1.007	1.790	2.795	4.025	5.480	7.16	9.06	11.18
7.3	0.11336	0.454	1.021	1.814	2.834	4.081	5.556	7.26	9.18	11.34
7.4	0.11491	0.460	1.035	1.839	2.873	4.137	5.632	7.36	9.31	11.49
7.5	0.11646	0.466	1.049	1.864	2.912	4.193	5.708	7.46	9.43	11.65
7.6	0.11801	0.472	1.063	1.888	2.950	4.249	5.784	7.56	9.56	11.80
7.7	0.11956	0.479	1.077	1.913	2.989	4.305	5.860	7.66	9.68	11.96
7.8	0.12111	0.485	1.091	1.938	3.028	4.361	5.936	7.76	9.81	12.11
7.9	0.12266	0.491	1.105	1.963	3.066	4.417	6.012	7.86	9.93	12.27
8.0	0.12424	0.497	1.118	1.988	3.105	4.472	6.089	7.96	10.06	12.42
8.1	0.12579	0.503	1.132	2.013	3.144	4.528	6.165	8.06	10.19	12.58
8.2	0.12734	0.509	1.146	2.038	3.183	4.584	6.241	8.16	10.32	12.73
8.3	0.12889	0.516	1.160	2.063	3.221	4.640	6.317	8.26	10.44	12.89
8.4	0.13044	0.522	1.174	2.087	3.260	4.696	6.393	8.36	10.57	13.04
8.5	0.13199	0.528	1.188	2.112	3.299	4.752	6.469	8.46	10.69	13.20
8.6	0.13354	0.534	1.202	2.137	3.338	4.808	6.545	8.55	10.81	13.35
8.7	0.13509	0.541	1.216	2.162	3.377	4.864	6.621	8.65	10.94	13.51
8.8	0.13664	0.547	1.230	2.187	3.416	4.920	6.697	8.75	11.07	13.66
8.9	0.13819	0.553	1.244	2.212	3.455	4.976	6.773	8.85	11.19	13.82
9.0	0.13977	0.559	1.258	2.237	3.493	5.031	6.850	8.95	11.32	13.98
9.1	0.14132	0.565	1.272	2.262	3.532	5.087	6.926	9.05	11.45	14.13
9.2	0.14287	0.571	1.286	2.287	3.571	5.143	7.002	9.15	11.58	14.29
9.3	0.14442	0.578	1.300	2.312	3.609	5.199	7.078	9.25	11.70	14.44
9.4	0.14597	0.584	1.314	2.336	3.648	5.255	7.154	9.35	11.83	14.60
9.5	0.14752	0.590	1.328	2.361	3.687	5.311	7.230	9.45	11.95	14.75
9.6	0.14907	0.596	1.342	2.386	3.726	5.367	7.306	9.55	12.08	14.91
9.7	0.15062	0.603	1.356	2.411	3.765	5.423	7.382	9.65	12.20	15.06
9.8	0.15217	0.609	1.370	2.435	3.804	5.479	7.458	9.75	12.33	15.22
9.9	0.15372	0.615	1.384	2.460	3.843	5.534	7.534	9.85	12.45	15.37
10.0	0.15530	0.621	1.398	2.485	3.882	5.589	7.612	9.94	12.58	15.53
11	0.17083	0.68	1.54	2.74	4.27	6.15	8.37	10.9	13.8	17.08
12	0.18636	0.74	1.68	2.99	4.66	6.71	9.13	11.9	15.1	18.64
13	0.20189	0.81	1.82	3.24	5.05	7.27	9.89	12.9	16.3	20.19
14	0.21742	0.87	1.96	3.49	5.44	7.83	10.65	13.9	17.6	21.74
15	0.23295	0.93	2.10	3.74	5.83	8.39	11.41	14.9	18.9	23.30
16	0.24848	0.99	2.24	3.98	6.21	8.95	12.17	15.9	20.2	24.85
17	0.26401	1.06	2.38	4.23	6.60	9.51	12.93	16.9	21.4	26.40
18	0.27954	1.12	2.52	4.48	6.99	10.07	13.69	17.9	22.7	27.95
19	0.29507	1.18	2.66	4.73	7.38	10.63	14.45	18.9	23.9	29.51
20	0.31060	1.24	2.80	4.97	7.76	11.18	15.22	19.9	25.2	31.06

*For head loss in metres, factor by 0.3.

The value of K for a given component is independent of friction factor or Reynolds number and is constant for all conditions of flow, including laminar flow. Equally, in theory at least, it should be the same for all sizes of a given component with similar geometry. In practice this is not so. Resistance coefficients are necessarily determined empirically, and can vary widely with pipe size. Some typical values are summarised in Table 2.

Equivalent length L/D

Equating velocity head loss derived from the D'Arcy formula with that derived from the resistance coefficient formula:

$$(f \times L/d)\,\frac{V^2}{2g} = K \times \frac{V^2}{2g}$$

It follows that:

$$K = f \times L/D$$

If K is a constant for all conditions of flow (but modified by geometric dissimilarities in different sizes, as above), the value of L/D for any given component must vary inversely with the change in friction factor for the different flow conditions. This rules out the effect of geometric dissimilarities in the sizes of valves, fittings, etc., of the same basic geometry. In other words, the equivalent length of L/D value for any particular valves, fittings, etc., is a constant for the same flow conditions, and is valid for all sizes of that particular component (see Figure 1 and Tables 3 and 4).

Flow coefficient C_v

For valves (and particularly control valves), it is often more convenient to express the capacity and flow characteristics in terms of a flow coefficient (C_v), defined as the flow of water at 60°F in US gal/min at a pressure drop of 1 lbf/in^2 across the valve.

Note: This definition of C_v based on the US gallon and pressure drop in lbf/in^2 is used both in countries employing Imperial units and those normally employing metric units. The metric 'equivalent' is called the flow factor, designated K_V and defined as the number of cubic metres per hour of water at 20°C which will flow with a pressure drop of 1 kg/cm^2 (1 bar). This gives somewhat different values for the same condition. Equivalents are:

$$K_V = 0.853\ C_v$$
$$C_v = 1.16\ K_V$$

Table 2. Resistance coefficient of fittings

Fitting	Pipe diameter—in (mm)									
	$^{3}/_{8}$ (10)	$^{1}/_{2}$ (12.5)	$^{3}/_{4}$ (20)	1 (25)	$1^{1}/_{4}$ (35)	$1^{1}/_{2}$ (40)	2 (50)	3 (75)	4 (100)	5 (125)
Integral pipe bend (turbulent flow) bend 3×D	Approximately 0.04 (all sizes)									
Integral pipe bend (turbulent flow) bend 4×D	Approximately 0.025 (all sizes)									
Integral pipe bend (laminar flow) bend 3×D	Approximately 0.1 (all sizes)									
Integral pipe bend (laminar flow) bend 5×D	Approximately 0.06 (all sizes)									
Integral pipe bend (laminar flow) bend 10×D	Approximately 0.04 (all sizes)									
Standard 90° elbow (screwed)	2.40	2.10	1.70	1.50	1.25	1.15	1.00	0.80	0.70	0.55
Standard 90° elbow (flanged)	—	—	—	0.45	—	0.40	0.38	0.34	0.32	0.30
Large-radius 90° elbow (screwed)	—	1.0	0.90	0.75	0.60	0.50	0.40	0.30	0.25	0.20
Large-radius 90° elbow (flanged)	—	—	—	0.40	—	0.34	0.30	0.25	0.21	0.18
Standard 45° elbow (screwed)	0.39	0.37	0.36	0.35	0.34	0.32	0.30	0.29	0.28	0.27
Standard 45° elbow (flanged)	0.37	0.35	0.34	0.33	0.32	0.31	0.29	0.28	0.27	0.26
Large-radius 45° elbow (screwed)	—	—	—	0.24	0.23	0.22	0.21	0.20	0.19	0.18
Large-radius 45° elbow (flanged)	—	—	—	0.22	0.21	0.21	0.20	0.18	0.18	0.17
Return bend 180° (screwed)	2.40	2.10	1.70	1.50	1.38	1.25	0.96	0.78	0.68	0.58
Return bend 180° (flanged)	—	—	—	0.43	0.40	0.37	0.34	0.32	0.30	0.28
Large-radius return bend 180° (screwed)	—	—	—	0.80	0.70	0.60	0.50	0.40	0.35	0.30
Large-radius return bend 180° (flanged)	—	—	—	0.42	0.39	0.36	0.30	0.25	0.22	0.20
T-line flow (screwed)	Approximately 0.9 (all sizes)									
T-line flow (flanged)	—	—	—	0.26	0.24	0.22	0.18	0.16	0.14	0.13
T-branch flow (screwed)	2.50	2.40	2.00	1.80	1.60	1.50	1.40	1.20	1.10	1.00
T-branch flow (flanged)	—	—	—	1.00	—	0.90	0.80	0.72	0.64	0.62
Screw-down valve (straight)	—	12.00	—	10.00	—	—	8.00	7.00	6.00	5.00
Screw-down valve (right-angle flow)	—	6.00	—	5.00	—	—	4.00	3.50	3.00	2.50
Gate valve (typical) (screwed)	—	—	—	—	0.23	0.20	0.17	0.15	0.13	0.11
Gate valve (typical) (flanged)	—	—	—	—	—	—	0.30	0.23	0.15	0.13
Gate valve: $^{1}/_{4}$-closed	0.8–0.2 this range									
Gate valve: $^{1}/_{2}$-closed	4.0–0.8 this range									
Gate valve: $^{3}/_{4}$-closed	16.0–2.0 this range									
Globe valve (screwed)	—	15.0	—	12.50	—	—	8.50	7.50	6.50	6.00
Globe valve (flanged)	—	—	—	12.50	—	—	8.50	7.50	6.50	6.00
Swing check valve (screwed)	—	5.0	—	3.00	—	—	2.00	2.00	2.00	2.00
Swing check valve (flanged)	Typically 2.0 (all sizes)									
Foot valve	Typically 0.8 (all sizes)									
Basket strainer	Typically 1.5–1.0 this range									

In terms of D'Arcy equation:

$$C_v = \frac{29.9d^2}{\sqrt{f \times L/D}} = \frac{29.9d^2}{\sqrt{K}}$$

The flow through a valve handling liquids with a reasonably similar viscosity to that of water can then be determined as:

$$Q = C_v\sqrt{\Delta P\frac{62.4}{w}} \quad = 7.9C_v\sqrt{\frac{\Delta P}{w}} \qquad \text{or} \ = C_v\sqrt{\frac{\Delta P}{sg}}$$

where w is the specific weight of the fluid (lb/ft^3)
sg is the specific gravity of the fluid
ΔP is the pressure drop across the valve.

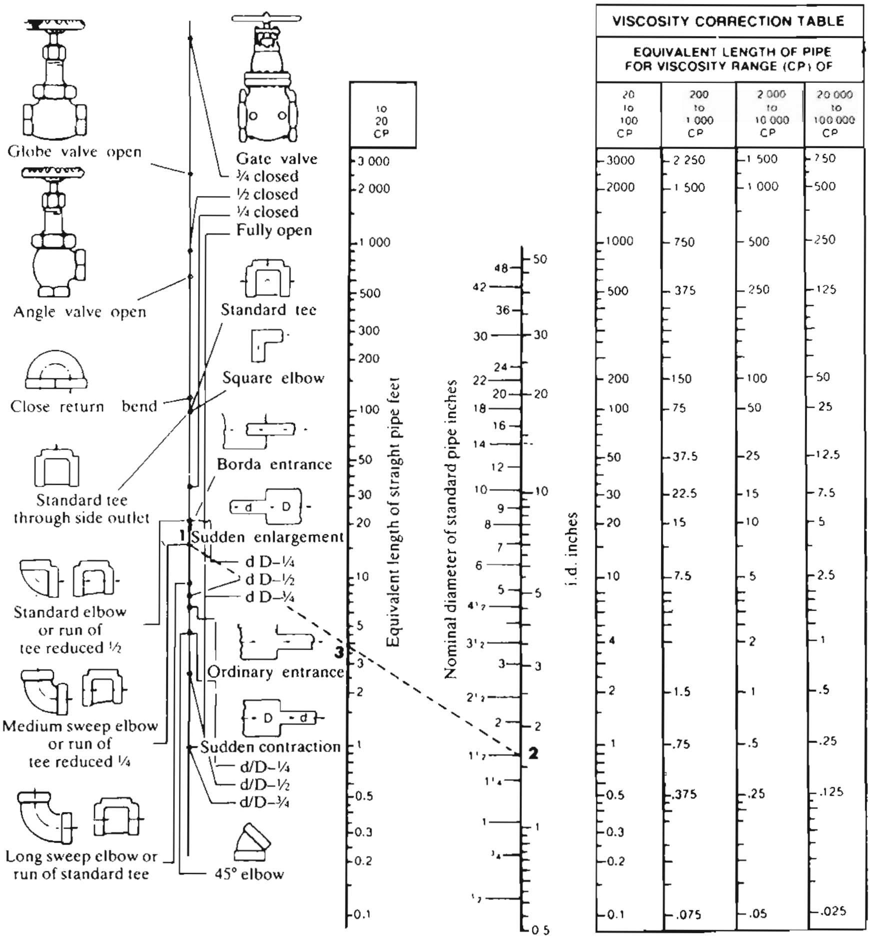


Figure 1. Friction loss in valves and fittings.

Table 3. Equivalent straight lengths of fittings in feet

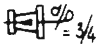
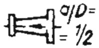
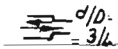
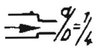
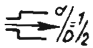
Fitting	Pipe diameter —in (mm)									
	6 (150)	8 (200)	10 (250)	12 (300)	14 (350)	16 (400)	18 (450)	20 (500)	24 (600)	36 (900)
Welded 90° L-bends R/D, where R = bend radius D = pipe diameter										
0.5	19	25	32	28	44	50	56	—	—	—
1.0	8	11	14	17	20	23	26	—	—	—
1.5	6	8	10	12	14	16	18	—	—	—
2.0	4	6	8	10	12	14	16	—	—	—
3.0	4	6	7	9	11	13	15	—	—	—
Standard 90° elbow (screwed)	12	16	—	—	—	—	—	—	—	—
Standard 90° elbow (flanged)	12	16	20	24	28	32	36	40	48	72
Large-radius 90° elbow (screwed)	—	—	—	—	—	—	—	—	—	—
Large-radius 90° elbow (flanged)	9	12	15	18	21	24	27	30	36	55
Standard 45° elbow (screwed)	—	—	—	—	—	—	—	—	—	—
Standard 45° elbow (flanged)	3	4	5	6	7	8	9	10	12	18
Large-radius 45° elbow (screwed)	—	—	—	—	—	—	—	—	—	—
Large-radius 45° elbow (flanged)	2	2.5	3.3	4	4.5	5.5	6	6.5	8	12
Return bend 180°	36	47	62	73	—	—	—	—	—	—
Large-radius return bend 180°	19	26	33	39	—	—	—	—	—	—
T-line flow	30	40	50	60	70	80	90	100	120	180
T-branch flow	150	200	250	300	350	400	440	480	570	850
Cast elbow (90°) standard	16	20	26	32	36	42	46	52	63	94
Cast elbow (90°) large radius	11	14	17	20	23	27	30	34	40	60
Screw-down valve (straight)	156	208	260	310	363	415	467	519	622	934
Screw-down valve (right-angled run)	75	100	125	150	175	200	225	250	300	450
Gate valve (typical) (flanged)	3.5	4.5	5.7	6.7	8.0	9.0	10.2	12.0	14.0	20.0
Gate valve: $^1/_4$-closed	19	26	33	39	—	—	—	—	—	—
Gate valve: $^1/_2$-closed	100	130	160	190	—	—	—	—	—	—
Gate valve: $^3/_4$-closed	400	540	700	800	—	—	—	—	—	—
Globe valve (flanged)	160	214	267	320	373	427	480	534	640	960
Swing check valve (screwed)	—	—	—	—	—	—	—	—	—	—
Swing check valve (flanged)	40	53	67	80	93	107	120	134	160	240
Foot valve (typical)	12	16	20	24	28	32	36	40	48	72
Basket strainer (typical)	11	14	16	18	21	24	30	35	—	—
Sudden enlargement										
d/D = $^1/_4$	16	20	26	31	36	42	—	—	—	—
d/D = $^1/_2$	11	15	18	22	25	29	—	—	—	—
d/D = $^3/_4$	3	4	6	7	8	10	—	—	—	—
Sudden contraction										
d/D = $^1/_2$	6	8	10	12	14	16	18	20	24	36
Entrance (typical)	9	12	15	18	21	24	27	30	—	—

d, smaller pipe diameter.

D, larger pipe diameter.

Table 4. Equivalent straight lengths of pipe in metres

		Fitting diameter—mm															
Fitting		1000	800	600	500	400	300	200	150	100	80	65	50	40	32	25	15
Gate valve		6	5	4	3	2.7	2.2	1.5	1	0.7	0.5	0.43	0.35	0.27	0.2	0.18	0.1
Non-return flap valve		110–200	90–170	70–130	60–110	50–90	35–70	25–50	20–35	13–25	10–20	8–15	6–12	5–9	4.7	3.6	2.4
Screw-down valve		300	250	200	160	130	100	70	50	35	28	22	17	13	10	8	5
Screw-down valve, right-angled		150	130	90	80	60	50	32	25	16	13	10	8	7	5	4	2.5
Bends and elbows		18	15	12	10	8	6	4	3	2	1.7	1.4	1	0.8	0.6	0.5	0.3
		25	20	15	13	10	8	5	4	2.8	2	1.8	1.5	1	0.8	0.7	0.4
		12	10	7	6	5	4	2.5	2	1.5	1	0.9	0.7	0.5	0.4	0.3	0.2
		30	25	18	15	13	10	6.5	5	3.2	2.5	2	1.8	1.4	1	0.8	0.5
		75	60	50	40	33	25	17	13	8	6.5	5.5	4	3.2	2.6	2	1.4

T fittings		100	80	60	50	40	30	20	15	10	8	7	5	4	3.2	2.7	1.7
		70	58	45	35	30	24	15	12	8	6	5	4	3	2.5	2	1.2
		18	15	12	10	8	6	4	3	2	1.7	1.4	1	0.8	0.6	0.5	0.3
Taper connectors	 $d/D = 3/4$	25	20	16	14	11	8	5.5	4	3	2.5	2	1.5	1.3	0.9	0.7	0.4
	 $d/D = 1/2$	30	25	20	16	13	10	7	5	3.5	2.8	2.2	1.7	1.4	1	0.8	0.5
Abrupt 90° bend		60	50	40	35	28	20	14	10	7	5.5	4.4	5	3	2	1.8	1
Abrupt changes of section	 $d/D = 3/4$	6	5	4	3.5	3	2	1.5	1	0.7	0.6	0.5	0.37	0.3	0.24	0.18	0.11
	 $d/D = 1/4$	13	10	8	7	5	4	3	2	1.5	1.2	0.9	0.7	0.55	0.45	0.35	0.2
	 $d/D = 1/2$	10	8	6	5	4	3.2	2.2	1.6	1.1	0.9	0.7	0.55	0.45	0.35	0.27	0.17
	 $d/D = 1/4$	30	25	20	16	13	10	7	5	3.5	2.8	2.2	1.7	1.4	1	0.8	0.5
	 $d/D = 1/2$	15	13	10	9	7	5.5	3.5	3	2	1.5	1.3	0.9	0.7	0.6	0.5	0.3
		15	13	10	8	7	5	3.5	2.5	1.8	1.5	1.2	0.9	0.7	0.5	0.4	0.25

The pressure drop across a valve can be determined from the same formulae rearranged:

$$\Delta P = \frac{w}{62.4}\left(\frac{Q}{C_v}\right)^2 = sg\left(\frac{Q}{C_v}\right)^2$$

Working formulae

These formulae are expressed in terms of the *resistance coefficient* (K) where

$$K = f\frac{L}{D}$$

Head loss through valves and fittings

$$H_L\ (ft) = 522\frac{KQ^2}{d^4}$$

where Q is in ft^3/sec
d is in inches

$$H_L\ (ft) = 0.00127\frac{KB^2}{d^4}$$

where B is in barrels (42 US gal/hr)
d is in inches

$$H_L\ (m) = 1{,}877{,}197\frac{KQ^2}{d^4}$$

where Q is in m^3/sec
d is in mm

$$H_L\ (m) = 6852\frac{KQ^2}{d^4}$$

where Q is in l/min
d is in mm

$$H_L\ (ft) = 0.00259\frac{KQ^2}{d^4}$$

where Q is in US gal/min
d is in inches

$$H_L\ (ft) = 0.00216\frac{KQ^2}{d^4}$$

where Q is in Imperial gal/min
d is in inches

$$H_L\ (ft) = 0.0000403\frac{KW^2\bar{V}^2}{d^4}$$

where W is in lb/bar
$\bar{V}$ (specific volume) is in ft^3/lb

Pressure drop through valves and fittings (using resistance coefficient K)

$$\Delta P\ (lbf/in^2) = 0.0001078k\rho V^2$$

where ρ is in lb/ft^3
V is in ft/sec

$$\Delta P\ (lbf/in^2) = 3.62\frac{K\rho Q^2}{d^4}$$

where ρ is in lb/ft^3
Q is in ft^3/sec
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.00000882 \frac{K\rho B^2}{d^4}$$

where ρ is in lb/ft^3
B is in barrels (42 US gal/hr)
d is in inches

$$\Delta P\ (\text{bar}) = 0.000000112\ K\rho V^2$$

where ρ is in tonnes/m^3
V is in m/sec

$$\Delta P\ (\text{bar}) = 0.1729 \frac{K\rho Q^2}{d^4}$$

where ρ is in tonnes/m^3
Q is in l/min
d is in mm

$$\Delta P\ (\text{lbf/in}^2) = 0.00000003\ K\rho V^2$$

where ρ is in lb/ft^3
V is in ft/min

$$\Delta P\ (\text{lbf/in}^2) = 0.000018 \frac{K\rho Q^2}{d^4}$$

where ρ is in lb/ft^3
Q is in US gal/min
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.000015 \frac{K\rho Q^2}{d^4}$$

where ρ is in lb/ft^3
Q is in Imperial gal/min
d is in inches

$$\Delta P\ (\text{lbf/in}^2) = 0.00000028 \frac{KW^2\bar{V}}{d^4}$$

where W is in lb/bar
$\bar{V}$ (specific volume) is in ft^3/lb

Pressure drop through valves and fittings (using flow coefficient C_v)

$$\Delta P\ (\text{lbf/in}^2) = \frac{\rho Q^2}{62.4\, C_v^2}$$

where Q is in lb/ft^3
ρ is in lb/ft^3
Q is in US gal/min

$$\Delta P\ (\text{bar}) = \frac{\rho Q^2}{223\, C_v}$$

where ρ is in tonnes/m^3
Q is in l/min

$$\Delta P\ (\text{lbf/in}^2) = \frac{\rho Q^2}{90\, C_v^2}$$

where ρ is in lb/ft^3
Q is in Imperial gal/min

Discharge through pipes and fittings

$$Q\ (\text{ft}^3/\text{s}) = 0.0438\,d^2\sqrt{\frac{H_L}{K}}$$

$$= 0.525\,d^2\sqrt{\frac{\Delta P}{K\rho}}$$

where d is in inches
H_L is in ft
ΔP is in lbf/in^2
ρ is in lb/ft^3

$$Q\ (\text{US gal/min}) = 19.65\,d^2\sqrt{\frac{H_L}{K}}$$

$$= 236\,d^2\sqrt{\frac{\Delta P}{K\rho}}$$

where d is in inches
H_L is in ft
ΔP is in lbf/in^2
ρ is in lb/ft^3

$$Q\ (\text{Imperial gal/min}) = 16.375\,d^2\sqrt{\frac{H_L}{K}}$$

$$= 197\,d^2\sqrt{\frac{\Delta P}{K\rho}}$$

where d is in inches
H_L is in ft
ΔP is in lbf/in^2
ρ is in lb/ft^3

$$Q\ (\text{lb/h}) = 197.6\rho d^2\sqrt{\frac{H_L}{K}}$$

$$= 1891\,d^2\sqrt{\frac{\Delta P}{K\rho}}$$

where d is in inches
H_L is in ft
ΔP is in lbf/in^2
ρ is in lb/ft^3

$$Q\ (\text{l/min}) = 0.1357\,d^2\sqrt{\frac{H_L}{K}}$$

$$= 6.06\,d^2\sqrt{\frac{\Delta P}{K\rho}}$$

where d is in mm
H_L is in m
ΔP is in bar
ρ is in tonnes/m^3

None of these formulae is strictly valid for predicting valve performance with viscous fluids or compressible fluids (i.e. air or gases).

Losses in bends

Losses in bends are difficult to evaluate other than on purely empirical lines. Attempts to rationalise resistance coefficients in terms of relative radius (ratio of bend radius to internal pipe diameter) are generally unsuccessful, but values are fairly well established for standard bends, including mitre bends.

Alternative data are presented in terms of equivalent length or L/D (see Tables 3 and 4 and Figure 1).

Contractions and enlargements

Flow through gradual changes in pipe sections (contractions or enlargements) can be analysed from first principles using the Bernoulli equation (Bernoulli constant), with the subsequent introduction of an empirical coefficient to take into account frictional losses introduced with a real fluid. In problems involving closed circuits, the potential head can normally be ignored in view of the inherently higher value of velocity and frictional head so that the original equation can be reduced to:

$$\frac{P}{\rho} + \frac{V^2}{2g} = \text{a constant}$$

This is often more convenient to use in the form:

$$V^2 + \frac{2Pg}{\rho} = \text{a constant}$$

Applying this condensed equation to steady frictionless flow along a length of horizontal pipe which contracts in cross-section, the following applies:

$$V_1^2 + \frac{2P_1g}{\rho} = V_2^2 + \frac{2P_2g}{\rho}$$

i.e.:

$$P_1 - P_2 = \frac{\rho}{2g}(V_2^2 - V_1^2)$$

This form of equation also shows that a steadily contracting section followed by a widening section can be used as a principle for flow velocity measurement by measuring the difference in pressure at two extreme sections, as in the venturi (Figure 2). If X expresses the ratio A_2/A_1, it follows that the volume of flow is the same at both sections that $V_1 = X \times V_2$. Hence, rearranging the equation above and rewriting $(P_1 - P_2)$ as ΔP:

$$V_1 = \sqrt{\frac{2g\Delta P}{\rho} \times \frac{x^2}{(1 - x^2)}}$$

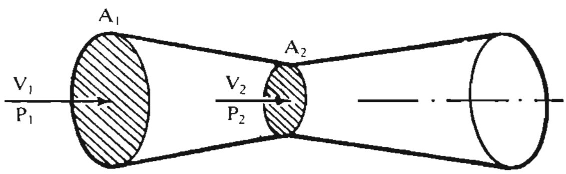


Figure 2.

This is the basic formula for the theoretical design of a venturi velocity meter. In practice, the formula is modified by the introduction of a calibration coefficient to take into account frictional losses, etc., neglected in the Bernoulli equation.

Sudden enlargements (Figure 3)

Again, flow conditions can be analysed from first principles, although for engineering calculations it is only necessary to determine the velocity head loss from the corresponding resistance coefficient K_1. This can be determined as:

$$K_1 = \left(1 - \frac{d_1^2}{d_2^2}\right)^2$$

where d_1 = i.d. of smaller pipe
d_2 = i.d. of larger pipe.

This formula is also quoted in the form:

$$K_1 = (1 - \beta^2)^2$$

when $\beta = d_1/d_2$.

Sudden contractions (Figure 4)

In this case, resistance coefficient is one half that for a sudden enlargement, viz:

$$K_2 = 0.5\left(1 - \frac{d_1^2}{d_2^2}\right)^2$$

or

$$K_2 = 0.5(1 - \beta^2)^2$$

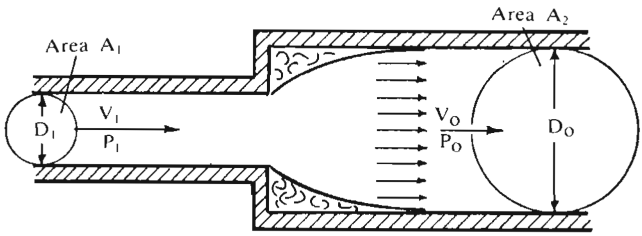


Figure 3.

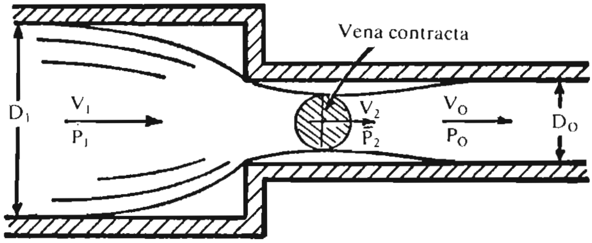


Figure 4.

Enlargement/contraction coefficients

$$\theta \leq 45^\circ \qquad C_e = 2.6 \sin\frac{\theta}{2}$$

$$45^\circ < \theta \leq 180^\circ \qquad C_e = 1$$

Coefficients for gradual contractions, derived by Crane, are:

$$\theta \leq 45^\circ \qquad C_c = 1.6 \sin\frac{\theta}{2}$$

$$45^\circ < \theta \leq 180^\circ \qquad C_c = \sqrt{\sin\frac{\theta}{2}}$$

where θ is the angle of divergence.

These formulae can also be expressed in terms of the larger pipe and have been extended to define resistance coefficients (K_3) for both sudden and gradual enlargements and contractions, viz:

Enlargement

$$45^\circ < \theta \leq 180^\circ \qquad K_3 = \frac{(1-\beta^2)^2}{\beta^4}$$

$$\theta \leq 45^\circ \qquad K_3 = \frac{2.6 \sin\frac{\theta}{2}(1-\beta^2)^2}{\beta^4}$$

Contraction

$$45^\circ < \theta \leq 180^\circ \qquad K_3 = \frac{0.5 \sin\sqrt{\frac{\theta}{2}}(1-\beta^2)}{\beta^4}$$

$$\theta \leq 45^\circ \qquad K_3 = \frac{0.8 \sin\frac{\theta}{2}(1-\beta^2)}{\beta^4}$$

Resistance coefficients and equivalent straight lengths applicable to changes of cross-section are given in Tables 5A, B and C.

Flow through orifices

Flow through a sharp-edged orifice is particularly significant as this has a number of practical applications, e.g. an orifice plate can be used as an indicating flow meter, though normally restricted in application of pipes of relatively large diameter, say over 50 mm (2 in).

Orifices are, in fact, principally used to meter rate of flow. They are also used to restrict flow or reduce pressure. With liquid flow, several orifices may be

Table 5A. Change of cross-section: equivalent straight lengths

Pipe size (d)														
mm	25	50	75	100	125	150	200	250	300	350	400	450	500	600
in	1	2	3	4	5	6	8	10	12	14	16	18	20	24
Expansion														
d/D = 0.25	3	6	8	11	14	16	20	26	31	36	42	48	55	65
0.5	2	4	6	7	9	12	15	18	22	25	29	34	46	52
0.25	5	1	2	2	3	3	4	6	7	8	10	12	15	20
Contraction														
d/D = 0.5	1	2	3	3	4	6	8	10	12	14	16	18	20	26

d, smaller pipe diameter.
D, larger pipe diameter.

Table 5B. Change of cross-section: resistance coefficient

Change of section	Ratio of smaller pipe diameter to larger pipe diameter (d/D)								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
10° taper	—	—	—	0.35	0.25	0.20	—	—	—
20° taper	—	—	—	0.15	0.12	0.10	—	—	—
Sudden expansion	2.0	—	—	—	—	—	—	—	0.15
Sudden contraction	0.5	0.45	0.45	0.45	0.45	0.4	0.3	0.2	—

Table 5C. Inlets and outlets: resistance coefficient

Change of section		Typical value
Inlet:	Abrupt	0.5
	Gradual	Not less than 0.5
	Projecting tube	1.0
Outlet:	Abrupt	1.0
	Gradual	Not less than 0.12

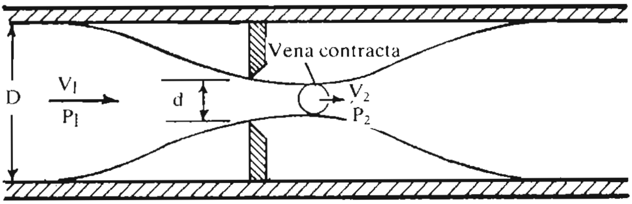


Figure 5.

used in sequence to reduce pressure in gradual steps, thus minimising the risk of cavitation occurring.

Simple analysis is restricted to streamlined flow, the flow pattern being of the form shown in Figure 5. The streamlines converge on approaching the orifice and continue to converge after passing through the orifice, reaching a minimum cross-sectional area, known as the vena contracta, downstream of the orifice before diverging again. For small circular orifices, the downstream position of the vena contracta is of the order of half the diameter of the orifice. At the vena contracta, all the streamlines are perpendicular to the plane of the orifice.

Strength of Pipes (Calculations)

In the case of homogeneous tubes (e.g. drawn tubes), a suitable pressure rating can be determined directly in terms of the D/t ratio by assuming a uniform distribution of stress through the tube wall, when:

$$P = \frac{2St}{D}$$

where D = outside diameter
t = wall thickness
P = internal pressure
s = material stress

This can be rendered in terms of a working pressure rating p_w where:

$$p_w = \frac{2S_{max}t}{D}$$

where S_{max} = the maximum permissible material stress for the material used. This is generally taken as one third the ultimate material stress (see Table 1).

Written as a solution for tube-wall thickness:

$$t = \frac{p_w D}{2S_{max}}$$

Provided the maximum material stress figure is taken within the limit of proportionality of the material, this simple formula is valid. It does not hold true for higher stress values, and thus will not accurately predict bursting pressures, e.g. using S_{ult} in place of S_{max}. It is also not valid where the ratio D/t is 16:1 or less, as stress is then no longer uniformly distributed through the wall thicknesses, but ranges from a maximum at the inner surface to a minimum at the outer surface. The simple formula is thus restricted to thin-walled tubes (D/t greater than 16:1).

It will over-estimate the pressure rating for thick-walled tubes (D/t 16:1 or less), and in such cases an alternative formula must be used. An alternative

Table 1. Maximum permissible stress for tube calculations (Minimum UTS divided by 3)

Material	Condition	P_{max} bar	P_{max} lbf/in²
Low carbon steel	As drawn	1280	18,300
	Drawn and polished*	2000	28,000
20-ton steel	As drawn	1000	15,000
Stainless (304) steel	Annealed	2350	33,300
	Half-hard	2800	40,000
	Hard	3500	50,000
Light alloy 61S-T6	As drawn	1000	15,000
Copper	Annealed	480	6800†
	Half-hard	630	9000†
	Hard	800	11,300†
Tungum	Annealed	1550	22,000
	Precipitation hardened	1550	22,000
Titanium 115/125		1300	18,500
Titanium 150/160		2100	30,000

*Cylinder tubes.
†Up to 65.5°C (150°F) only.

formula which can be applied in the case of thick-walled tubes and homogeneous metal pipes is:

$$S_{max} = P \times \frac{R_i^2}{R_O^2 - R_i^2} \times \left(\frac{R_O^2}{R_i^2} + 1\right)$$

where R_i = inner radius of tube
R_O = outer radius of tube.

Alternative formulae, written as a solution for wall thickness required, are:

$$t = \frac{D}{2}\left(\sqrt{\frac{S+P}{S-P}} - 1\right)$$

$$t = \frac{D}{2}\left(\sqrt{\frac{3S+P}{3S-4P}} - 1\right)$$

where S = S_{max} the corresponding value of P is p_w
S = S_{ult} the corresponding value of P is the bursting pressure.

Modified formulae are used in the case of non-homogeneous tubes and also for non-metallic tubes.

(i) In the case of welded tubes, a correction factor may be introduced or, alternatively, a higher divisor may be used to establish P_{max} from the ultimate tensile strength of the material. This is not necessarily the invariable rule, as welded tubes can have the same working strength as drawn tubes. Corrections may be applied to tubes with welded connections on a similar basis, however.

(ii) In the case of cast tubes, a nominal (and substantially lower) value may be adopted for P_{max}. Cast tubes are associated with older hydraulic systems and large pipe sizes, where pressure rating is established on empirical lines, permitting fairly large tolerances in wall thicknesses.

(iii) In the case of copper pipes and tubes intended for brazed or soldered connections, the standard thin-walled formula is de-rated:

$$p_w = \frac{2S_{max}t}{D - 0.8t}$$

(iv) In the case of metallic tubes intended for threaded connections, an allowance is made for the reduction in tube strength due to threading. The following formula can then be used for thin-walled tubes:

$$p_w = \frac{2S_{max}(t - C)}{D - 0.8(t - C)}$$

where C is taken as equal to the depth of thread cut, with a minimum value of 1.25 mm (0.05 in).

(v) In the case of plastic tubes, an allowance is made for the higher elastic moduli of such materials, when a suitable formula is:

$$p_w = \frac{2S_{max}t}{D - t}$$

Pipe-wall thickness according to British Standards

Two British Standards are applicable for the calculation of pipe-wall thickness:

BS 806: 1975	Specification for ferrous pipes and piping installations for and in connection with land boilers.
BS 3351: 1971	Specification for piping systems for petroleum refineries and petrochemical plants.

Design to BS 806

Minimum pipe-wall thickness where the outside diameter (D) is used as a basis for calculation.

Table 2. Design stresses for ferrous pipes (BS 3351)

Material	Notes	Values of S for metal temperatures in °C not exceeding*										
		−200 to +50	50	100	150	200	250 (N/mm^2)	300	350	400	425	450
BS 3601 HFS, CDS, steel 22	1		125.5	114.5	103.9	99.3	94.5	89.7	85.0	75.0	65.7	56.9
BS 3601 HFS, CDS, steel 27	1		154.5	141.3	127.5	121.7	114.2	110.0	104.0	87.7	75.8	62.0
API 5L Grade A, seamless; steels open hearth, electric furnace and basic oxygen			122.5	111.9	101.3	95.8	91.3	86.5	82.0	73.1	64.8	55.8
API 5L Grade B, seamless; sub-merged arc, spiral weld; steels open hearth, electric furnace and basic oxygen			153.0	139.8	126.5	119.0	113.8	108.5	102.5	89.3	75.5	62.0
BS 3602 CDS, HFS and ERW, steel 25	2		130.0	125.5	120.6	117.3	111.5	96.5	88.9	78.2	67.9	57.7
BS 3602 CDS, HFS and ERW, steel 27	2		154.5	143.0	132.0	127.0	119.0	110.0	104.2	89.7	75.9	62.3
BS 3602 EFW Grade 28 B	2, 3		123.5	116.5	109.6	104.0	95.2	90.4	87.0	73.8	62.0	50.3
BS 3603 HFS, CDS, steel 27 LT-30	4, 5		154.5	143.3	132.0	126.5	119.0					
BS 3603 HFS, CDS, steel 27 LT-50	4, 5		154.5	143.3	132.0	126.5	119.0					
BS 3603 HFS, CDS, steel 503 LT-100	4, 5		159.0	147.8	135.3	129.3	121.9					
BS 3604 620 HFS, CDS, steel 27			166.5	144.2	144.0	139.0	124.5	114.0	109.5	104.0	101.5	98.5
BS 3604 621 HFS, CDS, steel 27			144.5	144.2	144.0	139.0	124.5	115.5	110.5	106.0	103.8	100.0

*Intermediate values may be obtained by linear interpolation.
(1) Limited application (see Section 3 of BS 3351).
(2) The design stress values for pipe with a longitudinal or spiral weld seam incorporate weld-joint factors as follows: API 5L submerged arc welded: 1.0; API 5LS Grade B: 1.0; BS 3602 ERW with Appendix A: 1.0; BS 3602 EFW: 0.8.
(3) The values for BS 3602 EFW are based on material to BS 1501, 151 or 161, Grade 28B (certified or hot-tested). For temperatures over 400°C, 161 material should be used.
(4) The BS 3603 pipes listed are impact-tested and intended for use in low-temperature service.
(5) Up to 250°C to cover 'steaming out' and flexibility calculations.
(6) The figures in parentheses alongside the grades are equivalent AISI types. There is no AISI equivalent of 845T.

Values of S for metal temperatures in °C not exceeding*														
475	500	525	550	575	600	625 (N/mm^2)	650	675	700	725	750	775	800	825
47.5	35.8													
48.6	35.8													
39.3	24.8													
92.3	81.0	62.8	43.8	28.9	17.2									
92.8	81.2	64.3	49.3	34.2	24.4									

Table 2. Continued

Material	Notes	Values of S for metal temperatures in °C not exceeding* −200 to +50	50	100	150	200	250 (N/mm²)	300	350	400	425	450
BS 3604 622 HFS, CDS, steel 27			144.5	134.2	129.6	125.0	120.5	115.5	111.0	106.0	103.8	101.0
BS 3604 622 HFS, CDS, steel 31			164.0	164.0	164.0	164.0	156.5	150.5	144.0	137.2	134.0	130.2
BS 3604 660 CDS, HFS			172.0	172.0	167.0	164.0	156.5	150.5	144.0	137.2	134.0	130.2
BS 3604 625 HFS			137.0	131.0	125.5	119.5	114.0	108.5	103.0	97.0	92.8	86.5
BS 3605 Grade 801 (304)	6		126.0	113.5	103.5	95.0	87.5	81.9	76.3	71.7	68.9	66.8
BS 3605 Grade 811 (304H)	6											
BS 3605 Grade 801L (304L)	6	106.8		102.8	90.3	76.5	68.2	63.3	60.0	57.2	55.8	
BS 3605 Grade 822T (321)	6	129.3		127.5	117.0	108.5	105.5	102.8	102.0	101.3	100.5	99.0
BS 3605 Grade 832T (321H)												
BS 3605 Grade 845 (316) and Grade 845T and Grade 846 (317)	6	129.3		128.2	123.5	120.6	118.7	118.0	117.5	116.5	115.5	114.0
BS 3605 Grade 845L (316L)	6	107.5		106.2	100.0	83.4	79.3	74.3	66.9	62.3	60.7	58.6

*Intermediate values may be obtained by linear interpolation.

(1) Limited application (see Section 3 of BS 3351).

(2) The design stress values for pipe with a longitudinal or spiral weld seam incorporate weld-joint factors as follows: API 5L submerged arc welded: 1.0; API 5LS Grade B: 1.0; BS 3602 ERW with Appendix A: 1.0; BS 3602 EFW: 0.8.

(3) The values for BS 3602 EFW are based on material to BS 1501, 151 or 161, Grade 28B (certified or hot-tested). For temperatures over 400°C, 161 material should be used.

(4) The BS 3603 pipes listed are impact-tested and intended for use in low-temperature service.

(5) Up to 250°C to cover 'steaming out' and flexibility calculations.

(6) The figures in parentheses alongside the grades are equivalent AISI types. There is no AISI equivalent of 845T.

Values of S for metal temperatures in °C not exceeding*														
475	500	525	550	575	600	625 (N/mm^2)	650	675	700	725	750	775	800	825
92.5	82.8	63.8	47.5	36.2	26.8									
107.5	82.8	63.8	47.5	36.2	26.8									
127.5	115.6	79.9	53.7	36.9										
81.0	72.4	59.0	43.8	31.3	21.1									
64.8	63.3	61.3	59.7											
				55.8	48.0	37.6	30.7	22.4	17.6	13.4	10.3	7.9	6.2	4.5
97.5	96.0	94.0												
			91.7	88.5	75.8	51.3	34.5	25.5	19.6	15.2	11.8	9.3	7.6	6.5
111.5	106.0	99.3	90.3	79.3	67.6	56.2	46.6	36.8	28.3	22.0	17.6	14.4	11.8	9.6

$$t_{min} = \frac{Pd}{2fe - P}$$

where t_{min} = the minimum thickness (mm)
P = the internal design pressure (N/mm^2)
D = the outside diameter (mm)
d = the inside diameter (mm)
f = the maximum permissible design stress as specified in Table 2 (N/mm^2)
e = 1.0 for seamless and for electric resistance welded steel pipes and for electric fusion welded steel pipes complying with the requirements of BS 3602 in which the weld is fully radiographed or ultrasonically tested.
e = 0.95 for electric fusion welded pipes complying with the requirements of BS 3602.
e = 0.90 for pipes complying with the requirements of BS 3601 other than electric resistance welded pipes.

The value of t_{min} is the minimum thickness for straight pipes and provision shall be made for any minus tolerance. Manufacturing considerations may make it necessary for pipes thicker than this minimum to be used.

$$t = \frac{PD}{20S + P}$$

where t = internal pressure design thickness (mm)
P = internal design pressure (bar)
D = outside diameter of pipe (mm)
S = design stress (N/mm^2) (shall be obtained from and shall not exceed those given in Table 2 for the design temperatures indicated therein). Linear interpolation for intermediate design temperatures is permitted.

Pipes with t equal to or greater than D/4 require special consideration.

Mitred bends

Although smooth bends are more common, mitred bends are widely used in industry. Typical applications include large-diameter pipelines or ducting in chemical complexes, de-salination plants and nuclear power stations, where the manufacture of smooth bends may be either impractical or uneconomical.

To satisfy requirements within the high-pressure pipeline industry in the UK, the BSI has produced recommendations on the design and application of gusseted or mitred bends. This is published in Amendment No. AMD 3545 to

BS 806. A feature of the recommendations is that they are to be used in conjunction with BS 806 stress limits, and thus complement the BS 806 smooth bend design assessment procedure.

Finite element methods have been used to analyse the stresses in single mitre pipe bends to verify experimental results.

Tee junctions and intersections

Pipelines used in the power generation industry, either in conventional or nuclear power stations, are subjected to complex loading. The pipelines from the boiler to the turbine can weigh up to 400 tons, and have to be stressed as a structure loaded by its own weight, as well as a pressure vessel at high temperature. Tee branch pipe intersections are therefore subjected to both internal pressure and external moment loadings arising as a result of dead-weight, thermal expansion, cold pull or seismic effects.

The following design codes are used for the calculation of stresses and the determination of stress concentration and flexibility factors:

BS 806: 1975	Specification for ferrous pipes and piping installations for and in connection with land boilers.
BS 5500: 1982	Specification for unfired fusion welded pressure vessels.
ANSI/ASME B.31.1: 1980	Power piping
ASME 111: 1983	Boiler and pressure vessel code: nuclear power plant components.

Finite element analysis methods have been used to compare the results obtained from the different codes with experimental results, and these have been published by the Institution of Mechanical Engineers in the United Kingdom.

See also standards EN545, EN598 and ISO 25321 as well as British Standard Code of Practice for Pipelines 8010 Section 2.1.

Buried Pipes

In designing rigid buried pipelines, the determination of the external load due to backfill and surface loadings is conventionally based on the methods and formulae established by Marston and Spangler. These involve lengthy and tedious calculations but are readily adapted to CAD (computer-aided design). For general calculation, simplified tables are available, notably those by Young and Smith (Building Research Station Report, 1970), corresponding to normal practice with concrete pipes and incorporating corrections for differences in external diameter. The latter can be a significant feature in the case of cement pipes because of the smaller external diameter reducing the load which it has to carry. Further simplified tables have been computed on this basis.

Rigid metal pipes (e.g. cast iron) and pipes in flexible materials (e.g. reinforced glass fibre) need somewhat different treatment. The former can be laid at any depth with 75 to 600 mm (3 to 24 in) of cover under buildings; and with not less than 300 mm (12 in) under roads and yards subject to normal usage. Elsewhere in good ground, such pipes will only need extra protection where subject to special loadings or abuse. In the latter case, design is usually based on traditional empirical data; recommendations based on experience or derived from experimental data evaluating specific structural protection requirements. With flexible pipe materials, due allowance must be given to the diametrical deflection produced by soil load, e.g. using Spangler's formula.

Cement-buried pipelines

The following covers the use of simplified tables for the design of cement-buried pipelines. Metric units are employed throughout, consistent with current British and European practice. General assumptions are:

(i) Backfill density of 2000 kg/m^3 (125 lbf/ft^3)
(ii) Frictional values $K\mu$ between 0.13 and 0.19
(iii) $r_{sd}p$ values of 0.5 and 0.7 as indicated.

Appropriate conversions are:

1 mm	= 0.0394 in	1 kg/m^3	= 0.0624 lbf/ft^3
1 m^2	= 10.76 ft^2	1 N	= 0.2248 lbf
1 kg	= 2.2046 lb	1 N/mm^2	= 1 MPa = 145 lbf/in^2
1 m	= 3.2008 ft	1 kN/m	= 68.52 lb/ft

Notation

B_c Outside diameter of pipe barrel.
B_d Trench width, measured at the level of the top of the pipe.
F_m Marston bedding factor, the value of which depends on the bedding method employed.
F_s Design factor of safety (1.25).
H The height of soil cover, measured from the top of the pipe barrel to the ground or road surface.
Y Soil density in kg/m^3 (2000).
$K\mu$ Product of the Rankine coefficient for the soil (K) and coefficient of internal friction of the soil (μ).
rsdp Product of p, the proportion of the pipe diameter that projects above the bedding, and rsd, the 'settlement ratio'.
W_e Total vertical external load imposed on the buried pipe (kgf/m).
W_T Minimum ultimate crush load of asbestos cement pipe (kgf/m).

Application of the tables

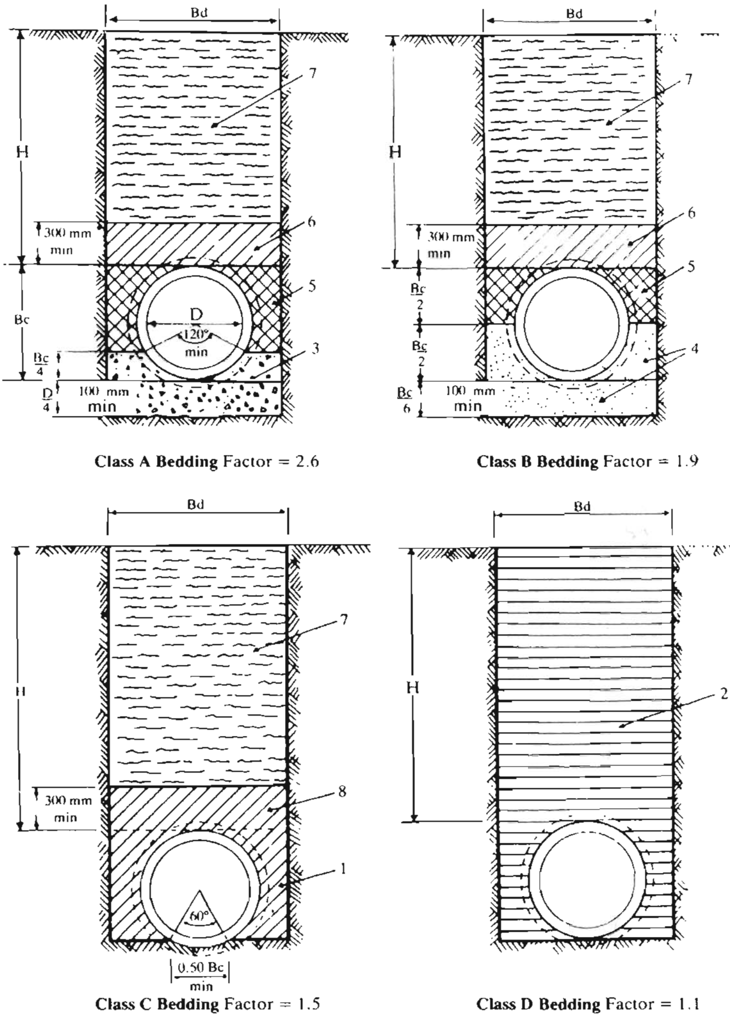
Tables 1–3 are based on assumptions that will be safe for a wide range of site conditions. The equivalent water loads are included in these tables. By separating the concentrated surcharge loads, Table 4 enables the designer to vary the calculation of the backfill load to suit individual circumstances, but here the water load must be added from Table 5 for pipes of 600 mm and over.

Pipes laid under verges should be designed for the full-vehicle loads. Buried pipes must not be exposed to excessive loads from heavy equipment during construction. Choice of the appropriate loading category for a given location is a matter for the engineer's judgement, with due regard to possible future changes. The tables are not appropriate for flexible pipes (pitch fibre, plastics, steel, etc.), nor for rigid pipes supported on piles.

Design method

For safe design, the minimum ultimate crush load (W_r) which the pipe is designed to withstand must be greater than or equal to the computed external load (W_e multiplied by a factor of safety of 1.25 and divided by the bedding factor F_m.

$$W_T \geq \frac{1.25W_e}{F_m}$$



1. Lightly compacted fill.
2. Loose fill.
3. 20.5 N/mm^2 at 28 days concrete well-packed under pipe.
4. Selected granular material well-tamped under and alongside the pipe.
5. Selected material well-tamped by hand in 75–150 mm layers.
6. Selected material lightly tamped by hand in 150 mm layers.
7. Normal fill.
8. Lightly compacted by hand.

Bd = Width of trench at crown of pipe.
H = Depth of cover over crown of pipe.
Bc = Outside diameter of pipe.

Bedding factor

Bedding class	Type of bedding	F_m
A (RC)	120° reinforced concrete cradle	3.4
A (Plain)	120° plain concrete cradle	2.6
B	360° granular bedding	2.2
B	120° granular bedding	1.9
C	Hand-shaped trench bottom	1.5
D	Hand-trimmed flat bottom	1.1

Figure 1. Bedding factors for pipes in trenches.

Table 1. (Metric units, kgf/m): main roads

Nominal diameter (mm)	Outside diameter (mm)	Assumed trench width (m)	W_e $H =$ 0.9	1.2	1.5	1.8	2.4	3.0	4.6	6.1	7.6
100	125	—	1432	1352	1326	1332	1408	1535	2003	2202	2212
150	182	0.60	2047	1946	1913	1926	2040	2228	2743	2829	2881
175	208	0.60	2328	2216	2181	2197	2328	2543	2784	2856	2899
200	232	0.70	2587	2465	2428	2446	2593	2834	3342	3500	3565
225	260	0.70	2888	2756	2715	2737	2903	3174	3385	3528	3585
250	288	0.70	3190	3046	3003	3028	3212	3288	3429	3556	3604
300	339	0.75	3738	3574	3526	3556	3754	3841	4058	4257	4369
375	423	1.00	4585	4368	4292	4312	4547	4951	5930	6451	6797
450	504	1.05	5443	5190	5101	5127	5591	5894	6666	7272	7733
525	587	1.15	6320	6031	5930	5961	6291	6672	7416	8126	8680
600	671	1.20	7415	7089	6977	7012	7273	7546	8367	9190	9848
675	756	1.35	8372	8009	7882	7924	8352	8743	9819	10,935	11,797
750	837	1.45	9282	8883	8745	8793	9082	9479	10,676	11,876	12,913
825	921	1.50	10,230	9796	9646	9650	9826	10,223	11,507	12,831	14,041
900	1007	1.60	11,220	10,747	10,526	10,424	10,586	11,013	12,371	13,907	15,190
975	1075	1.85	12,000	11,498	11,324	11,387	12,000	12,613	14,574	16,545	18,307
1050	1157	1.90	12,950	12,405	12,220	12,290	12,750	13,380	15,365	17,520	19,450

Table 2. (Metric units, kgf/m): light roads

Nominal diameter (mm)	Outside diameter (mm)	Assumed trench width (m)	W_e $H =$ 0.9	1.2	1.5	1.8	2.4	3.0	4.6	6.1	7.6
100	125	—	1335	1122	1053	1052	1152	1316	1863	2105	2144
150	182	0.60	1850	1585	1503	1512	1667	1909	2540	2690	2782
175	208	0.60	2085	1796	1708	1721	1901	2180	2552	2697	2785
200	232	0.70	2302	1991	1897	1914	2117	2429	3083	3324	3440
225	260	0.70	2554	2217	2117	2139	2369	2719	3096	3331	3445
250	288	0.70	2806	2444	2337	2364	2621	2784	3109	3338	3449
300	339	0.75	3265	2816	2737	2772	3056	3248	3683	4002	4188
375	423	1.00	3966	3458	3301	3330	3677	4213	5462	6134	6572
450	504	1.05	4681	4095	3915	3955	4552	5014	6110	6895	7465
525	587	1.15	5412	4745	4544	4593	5082	5647	6768	7687	8370
600	671	1.20	6360	5612	5388	5447	5890	6375	7628	8689	9493
675	756	1.35	7166	6837	6088	6158	6793	7424	8987	10,370	11,400
750	837	1.45	7934	7026	6756	6836	7357	8019	9755	11,250	12,470
825	921	1.50	8737	7746	7455	7495	7927	8617	10,494	12,150	13,550
900	1007	1.60	9576	8502	8128	8067	8510	9257	11,264	13,160	14,660
975	1075	1.85	10,236	9097	8762	8870	9782	10,738	13,366	15,750	17,740
1050	1157	1.90	11,037	9818	9460	9579	10,352	11,364	14,094	16,660	18,840

True values of F_m are considerably dependent upon standards of workmanship and good compaction of the side fill. Those tabulated assume properly maintained and supervised standards.

Method of use for Tables 1–3

Select the table appropriate to the type of surface loading. From the pipe diameter and the cover depth, the external load is read off directly. Use the given bedding factor to calculate the minimum value of W_r, and find the class of pipe required from Table 4. If a pipe of sufficient strength is not available a better class of bedding may be specified.

In general, the cover depth will vary along the main, and the pipe strength and bedding class must be selected to meet the maximum cover-depth condition. With pipelines having considerable variation in cover depths, it may be worth specifying different classes of pipe and/or bedding for different sections of the main. Alternatively, the tables may be used to find the limits of permissible cover depths for the different pipe and bedding classes available.

Method of use for Table 4

Where the trench conditions and backfill density deviate from the norm, Table 4 may be used to obtain the most economical design.

Table 3. (Metric units, kgf/m): fields, etc.

Nominal diameter (mm)	Outside diameter (mm)	Assumed trench width (m)	W_e $H =$ 0.9	1.2	1.5	1.8	2.4	3.0	4.6	6.1	7.6
100	125	—	809	792	825	886	1054	1253	1837	2091	2133
150	182	0.60	1174	1150	1198	1288	1532	1821	2502	2670	2769
175	208	0.60	1341	1313	1368	1470	1750	2080	2509	2674	2771
200	232	0.70	1494	1463	1525	1639	1951	2319	3035	3297	3424
225	260	0.70	1673	1638	1707	1835	2185	2598	3043	3302	3427
250	288	0.70	1852	1813	1889	2031	2418	2650	3051	3306	3429
300	339	0.75	2176	2131	2221	2388	2820	3090	3615	3964	4164
375	423	1.00	2655	2578	2671	2859	3389	4021	5378	6087	6543
450	504	1.05	3157	3066	3177	3401	4212	4786	6011	6840	7431
525	587	1.15	3670	3564	3693	3954	4689	5384	6654	7623	8329
600	671	1.20	4397	4276	4423	4722	5442	6076	7497	8616	9448
675	756	1.35	4979	4843	5009	5345	6292	7089	8841	10,290	11,346
750	837	1.45	5535	5384	5567	5940	6803	7649	9593	11,163	12,414
825	921	1.50	6117	5950	6152	6513	7320	8210	10,316	12,046	13,492
900	1007	1.60	6729	6547	6709	6998	7848	8814	11,070	13,049	14,591
975	1075	1.85	7211	7016	7252	7731	9077	10,266	13,159	15,630	17,668
1050	1157	1.90	7795	7586	7839	8355	9605	10,856	13,872	16,533	18,766

The method of use is:

Step 1 Knowing the pipe diameter and cover depth, obtain the vehicle load and the wide-trench load.
Step 2 Knowing the trench width, read off also the narrow-trench load. Adopt the lesser of the two backfill loads.
Step 3 If the soil density γ differs from 2000 kg/m^3 (125 lb/ft^3), correct the backfill load by $\gamma/2000$ ($\gamma/125$).
Step 4 Add the backfill. vehicle loads and equivalent water load for pipes over 600 mm to obtain W_e.

With W_e determined, the class of pipe and bedding required may be worked out as in Method of use for Tables 1–3 above.

Example

Determine the strength classes and bedding required through a length of 600 mm nominal diameter asbestos–cement pipeline laid under fields at cover depth ranging from 1.2 to 6.4 m (narrow-trench conditions).

Consider the use of Class B ($F_m = 1.9$) bedding or, if ground conditions permit, a Class D ($F_m = 1.1$) bedding. The maximum permissible external load can now be found by using the equation shown under Design method:

$$\text{i.e. } W_e \geq \frac{1.1 \times W_T}{1.25} = 0.88W_T$$

$$\text{or } W_e \geq \frac{1.9 \times W_T}{1.25} = 1.52W_T$$

The values of W_T for the different classes of pipe can be found in Table 4.

W_T for 600 mm diameter Class L pipes = 4300 kgf/m.
W_T for 600 mm diameter Class M pipes = 5800 kgf/m.

The permissible depths of cover can now be found from Table 4.

Table 4A.

	W_e (kgf/m)		Depth of cover (m)	
Class	$F_m = 1.1$	$F_m = 1.9$	$F_m = 1.1$	$F_m = 1.9$
L	3784	6536	—	0.6–3.7
M	5104	8816	0.6–2.1	0.6–7.0

A 600 mm Class L pipe, on a Class B bed, can be laid from 0.6 to 3.4 m; from 3.4 to 6.4 m, a Class M pipe laid on a Class B bed would be needed.

If ground conditions permit, a 600 mm Class M pipe can be flat bedded from 0.6 to 2.1 m, and laid on a Class B bed from 2.1 to 6.4 m.

Table 4. Class of pipe required

Nominal internal diameter (mm)	Outside diameter B_c (mm)	Trench width B_d (m)	Type of load	Total design load W_e in kilograms per metre of pipe length for cover depth *H* in metres									
				H= 0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0	3.4
100	125	—	Narrow										
			Wide (0.7)	233	350	468	585	703	821	938	1056	1173	1330
			Main road		1083	887	743	631	543	471	413	364	311
			Light road		985	656	468	350	271	215	174	144	114
			Fields, etc.	689	459	324	239	183	144	116	95	79	63
150	182	0.60	Narrow				1370	1550	1710	1840	1960	2080	2170
			Wide (0.7)	337	508	680	851	1022	1193	1364	1536	1707	1935
			Main road		1541	1268	1065	906	780	677	593	523	446
			Light road		1343	907	653	491	381	303	246	203	161
			Fields, etc.	1002	666	470	347	265	209	168	137	114	91
175	208	0.60	Narrow				1370	1550	1710	1840	1960	2080	2170
			Wide (0.7)	384	580	776	971	1167	1363	1559	1754	1950	2211
			Main road		1750	1442	1211	1031	888	771	675	595	508
			Light road		1506	1021	737	555	431	343	279	231	183
			Fields, etc.	1145	761	537	397	303	238	191	157	130	104
200	232	0.70	Narrow				1590	1810	2010	2190	2340	2470	2600
			Wide (0.7)	428	646	864	1083	1301	1519	1737	1956	2174	2465
			Main road		1942	1603	1347	1147	987	858	751	662	564
			Light road		1656	1127	815	614	478	381	309	256	203
			Fields, etc.	1276	848	599	442	338	265	213	174	145	116
225	260	0.70	Narrow				1590	1810	2010	2190	2340	2470	2600
			Wide (0.7)	478	723	968	1212	1457	1701	1946	2191	2435	2761
			Main road		2167	1790	1505	1282	1104	959	840	740	632
			Light road		1832	1251	906	683	532	424	345	285	226
			Fields, etc.	1430	950	671	495	378	297	239	195	162	130
250	288	0.70	Narrow				1590	1810	2010	2190	2340	2470	2600
			Wide (0.7)	529	799	1070	1341	1612	1883	2154	2425	2696	3058
			Main road		2392	1977	1663	1417	1220	1060	928	818	699
			Light road		2008	1374	997	752	586	467	380	314	249
			Fields, etc.	1584	1052	743	548	419	329	264	216	180	143
300	339	0.75	Narrow				1810	2060	2300	2510	2700	2880	3030
			Wide (0.7)	619	938	1257	1576	1895	2214	2533	2852	3171	3596
			Main road		2801	2319	1952	1663	1432	1244	1089	961	820
			Light road		2328	1559	1162	878	684	546	445	368	292
			Fields, etc.	1863	1238	873	645	492	387	310	254	211	168
375	423	1.00	Narrow				2480	2870	3230	3570	3870	4150	4420
			Wide (0.5)	733	1111	1489	1867	2246	2624	3002	3380	3758	4262
			Main road		3476	2881	2426	2068	1781	1547	1355	1195	1020
			Light road		2855	1969	1434	1086	847	676	551	456	362
			Fields, etc.	2324	1544	1089	804	614	482	387	316	263	210
450	504	1.05	Narrow				2710	3140	3540	3910	4260	4580	4880
			Wide (0.5)	868	1318	1769	2220	2670	3120	3751	4022	4473	5073
			Main road		4126	3423	2884	2459	2118	1840	1611	1421	1213
			Light road		3363	2327	1697	1286	1003	801	653	541	429
			Fields, etc.	2769	1839	1297	957	731	574	461	377	313	249
525	587	1.15	Narrow				2940	3420	3860	4260	4650	5020	5350
			Wide (0.5)	1004	1529	2054	2579	3104	3628	4153	4678	5203	5902
			Main road		4792	3979	3353	2859	2463	2140	1874	1652	1410
			Light road		3883	2692	1966	1490	1164	930	758	627	498
			Fields, etc.	3224	2141	1510	1114	851	668	536	438	364	290
600	671	1.20	Narrow				3170	3690	4180	4620	5050	5450	5830
			Wide (0.5)	1141	1740	2340	2940	3540	4140	4740	5340	5940	6740
			Main road		5467	4540	3827	3264	2812	2443	2139	1886	1610
			Light road		4410	3063	2238	1698	1326	1060	864	715	568
			Fields, etc.	3685	2447	1726	1273	972	763	612	501	416	332
675	756	1.35	Narrow				3630	4240	4800	5350	5850	6350	6800
			Wide (0.5)	1277	1953	2629	3304	3980	4956	5332	6008	6684	7585
			Main road		6149	5110	4308	3674	3165	2750	2408	2123	1812
			Light road		4943	3438	2514	1908	1490	1191	971	804	639
			Fields, etc.	4151	2757	1944	1434	1095	859	690	564	469	374
750	837	1.45	Narrow				3850	4500	5120	5710	6270	6800	7280
			Wide (0.5)	1405	2153	2901	3650	4398	5146	5895	6642	7391	8389
			Main road		6799	5652	4765	4065	3501	3042	2664	2349	2005
			Light road		5451	3795	2776	2108	1647	1317	1073	889	706
			Fields, etc.	4596	3052	2152	1587	1212	951	763	624	519	413